EXAMENSARBETE

SBW Feedback

Design of feedback system for increased usability in monostable SBW shifters



Tanya Alvarez Cabrera

Civilingenjör, Teknisk design
2017

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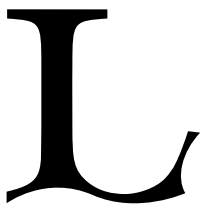
SBW Feedback

-Design of feedback system for increased usability in monostable SBW shifters

Tanya **Alvarez Cabrera**

2017

SUPERVISOR: Camilla **Grane**
REVIEWER: Peter **Törlind**
EXAMINER: Åsa **Wikberg Nilsson**



MSc in INDUSTRIAL DESIGN ENGINEERING
Department of Business Administration, Technology and Social Sciences
Luleå University of Technology

CIVILINGENJÖR I TEKNISK DESIGN
Master of Science in Industrial Design Engineering

SBW Feedback
Design of feedback system for increased usability in monostable SBW shifters

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Acknowledgement

The following thesis describes the design and evaluation process of a feedback system to increase the usability and safety of monostable shift-by-wire shifters. It is part of my master thesis for the Industrial Engineering program with the major in Product Design, at the Luleå University of Technology (LTU).

I would first like to thank my supervisor Dr. Camilla Grane, Senior Lecturer in Engineering Psychology at LTU. This project would not have happened without her support, guidance and honest dedication. She consistently made sure that I would be in charge of my own work and made my own decisions, but guided me and steered me in the right direction whenever it was needed.

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Luleå 12th of January, 2017



Tanya Alvarez Cabrera

Abstract

Electromechanical shift-by-wire car transmission systems make way for new innovative shifter designs such as monostable shifters that spring back to a starting position after a gear has been chosen. Unfortunately the radical change in the communication between the user and shifter has resulted in accidents due to incorrect gear selection. The inadequate usability of the monostable shifters can be attributed to the feedback it provided to users.

The aim of the thesis project was to develop a feedback concept that would improve the usability of a monostable shifter and to study if auditory feedback could be introduced to vehicle systems. By implementing design theory, benchmarking and various creativity methods five concepts with feedbacks in different modalities were developed. The concepts were evaluated in a usability study that involved 25 test participants. A review of the observations from the usability study along with an analysis of the interviews and collected data resulted in the final VRA concept. VRA was a multimodal concept with permanent visual feedback and optional auditory feedback. The shifting pattern was shown on the instrument cluster where the active gear was highlighted through light intensity, color and shape contrasts. The solid blue block, within which the abbreviation for the active gear was displayed, could be seen in the peripheral view. It was perceived as calm and helped the users navigate the shifter. A female machine voice that had a Swedish pronunciation was chosen as the auditory feedback to accompany the “P”, “R” and “D” gear selections. None of the users were indifferent towards the auditory feedback, some perceived it as caring while others found it annoying. Since the analysis did not indicate that auditory feedback was crucial, the VRA concept included an option to turn on or off the sound.

Monostable shifters behave differently compared to traditional polystable shifters, therefore with the changes in the physical movement the communication must also be reviewed. It is recommended to include the shifting pattern on the instrument cluster together with monostable shifters, as it makes up for the loss of the visual and haptic information from the physical shifter. Although the usability study showed that auditory feedback was not necessary, improvements were observed among people who favored it. The auditory information would most likely be better received if earcons were implemented instead of speech.

KEYWORDS: shift-by-wire, usability, shifter, feedback design, industrial design, multimodality.

Sammanfattning

Elektromekaniska shift-by-wire växlingssystem tillåter nya innovativa designlösningar så som monostabila växelväljare som fjädrar tillbaka till en ursprungsposition efter varje växelval. Dessvärre har den drastiska förändringen i kommunikationen mellan användarna och systemet resulterat i missförstånd och olyckor till följd av fel växelval. Den bristfälliga användbarheten av monostabila växelväljare kan härledas till feedbacken mellan förarens handling och resultat.

Syftet med examensarbetet var att utveckla feedbackkoncept som förbättrade användbarheten av en monostabil växelväljare samt att studera om auditiv feedback kunde introduceras i förarsystem. Genom att implementera kunskaperna inom designteori, observationer från benchmarking samt diverse kreativa metoder kunde fem olika koncept presenteras. Koncepten utvärderades i en användbarhetsstudie med 25 testdeltagare. En genomgång av observationerna från användbarhetsstudien, intervjusvaren samt dataanalysen resulterade i det slutgiltiga konceptet VRA. VRA var ett multimodalt koncept som innehöll permanent visuell feedback samt valbar auditiv feedback. Växlingsschemat visades på instrumentpanelen där den aktiva växeln markerades med hjälp av kontraster i ljusintensitet, färg och form. Förkortningarna på aktiva växel visades inuti en solid blå rektangel som var synlig i periferin. Färgen uppfattades som lugn och den starka kontrasten hjälpte personerna att navigera växelväljaren. Den auditiva feedbacken var i form av en kvinnlig maskinröst med svenskt uttal som kompletterade ”P”, ”R” och ”D” växellägena. Inga testpersoner var likgiltiga till auditiv information, vissa beskrev den som omtänksam medan andra ansåg att den var irriterande. Eftersom analysen indikerade att auditiv feedback inte var kritisk var den auditiva feedbacken valbar i VRA.

Monostabila växelväljare skiljer sig drastiskt från polystabila, därför bör kommunikationen mellan systemet och människan ses över i och med de fysiska förändringarna. Växlingsschemat bör inkluderas i instrumentpanelen i samband med att monostabila växelväljare används. Det kan nämligen kompensera bristen på den visuella och haptiska feedbacken från den fysiska väljaren. Trots att analysen visade på att auditiv feedbak inte var nödvändig kunde förbättringar observeras bland personer som tyckte om det. Den auditiva informationen skulle mest troligt få ett bättre bemötande om earcons användes istället för tal.

NYCKELORD: shift-by-wire, användbarhet, växel, återkoppling, teknisk design, multimodalitet.

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List of abbreviations

A	Auditory concept
AT	Automatic transmission
C	Control concept
D	Drive
M1	Manual 1
M2	Manual 2
M3	Manual 3
MT	Manual transmission
N	Neutral
P	Park
R	Reverse
SBW	Shift-by-wire
TL	Test leader
TP	Test person
V	Visual concept
VA	Visual and auditory concept
VT	Visual concept, temporary

1 Introduction

Master thesis in Industrial Design Engineering with the profile Product Design at Luleå University of Technology (LTU). The thesis focused on the usability aspect and communication between a monostable gear shifter and drivers.

1.1 BACKGROUND

The vehicle industry is evolving together with the technological development resulting in new in-vehicle systems and changes to existing controllers. Gear shifters are some of the elements that are currently going through changes as shift-by-wire (SBW) technology is implemented in automotive transmissions (Kelling & Leteinturier, 2003). The SBW electromechanical linkage between the gear and shifter means that traditional designs of shifters are no longer required, leading to new designs in existing cars as well as concept cars. One of those novelties are monostable shifters, gear selectors with a spring return, which can improve the ergonomic comfort, free space and contribute to a vehicle brand identity (Lindner & Tille, 2010).

If new in-vehicle designs are not thoroughly tested in terms of usability, they can become a threat to society by diverting the attention of the driver from the road (Wierwille, 1993). In 2015 accidents resulting from SBW shifters were reported (Donohue, 2015, 22 February) and research was initiated to investigate gear shifter usability issues. The division of Human Work Science at Luleå University of Technology (LTU) looked into this and presented parts of their results in December 2015 (Grane, 2015). Their studies showed that monostable gear selectors required more time and attention than traditional polystable gear selectors that remained in the chosen position.

Monostable shifters decrease the visual and haptic feedback as the shifter does not physically stay in the chosen gear. This removes some of the redundancy that is provided by polystable shifters, where feedback is given as a visual cue on the instrument cluster and through the position of the shifter. The design of the feedback in monostable shifters must therefore also be altered and new ways of redundancy might improve the usability. Auditory cues can be used in this case to complement or be redundant to the visual information on the cluster thus reinforcing the information (Hereford & Winn, 1994).

The studies conducted at LTU did not include any active visual or auditory feedback in the monostable or polystable shifters. A natural question that arose from the results was how the usability of monostable gear selectors could be improved so that the task completion time, amount of errors and work load would be reduced while the user satisfaction increased. A hypothesis was that feedback might improve the usability.

1.2 STAKEHOLDERS

The project was part of a research conducted by the division of Human Work Science at LTU, specifically Dr. Camilla Grane and PhD student Sanna Lohilahti Bladfält at Engineering Psychology. It was a continuation of a collaboration between Engineering Psychology with Kongsberg Automotive and Volvo Car Corporation; however this project was independent from that collaboration.

1.3 OBJECTIVE AND AIMS

The aim of the thesis was to study if the usability of monostable shifters could be improved through feedback. The difference in the feedback focused on two modalities, visual and auditory. The objective was to design a feedback concept that improved the usability of a specific monostable SBW shifter. The usability was investigated in terms of efficiency, effectiveness and user satisfaction of different concepts.

The research questions of the thesis work were:

- Is visual, auditory or multimodal feedback better than no feedback?
- What type of feedback is best in terms of efficiency, effectiveness and user satisfaction?
- How does the permanent representation of visual stimuli feedback affect the performance compared to temporary feedback?

1.4 PROJECT SCOPE

The physical design of the gear selectors couldn't be altered and the study could only focus on one specific design of a monostable shifter. There were no comparisons between monostable and polystable shifters. The design of a prototype set boundaries for what could be tested regarding forms of feedback. Haptic feedback was not tested. The laws and standards surrounding the automatic transmissions shifts were followed, unless a certain concept was used only for research purposes.

There was no correlation between the shifter and the Lane Change Test (LCT) driving simulator that was used for the usability tests. The time frame for the project did not allow usability tests of more than five different concepts. The variables that made the concepts different could not be too many in order to pinpoint why the results differ. Because of this only a small selection of possible feedback systems were tested, thus there might be other solutions that would yield better results.

1.5 THESIS OUTLINE

The thesis includes the information that was relevant for the development and evaluation of SBW feedback designs, the work process, the results and lastly an analysis of the results.

Section 2 explains what SBW is, the current regulations for feedback designs, the difference between poly- and monostable shifters, as well as problems that have been registered due to monostable shifter designs. The theoretical framework in section 3 describes usability, information processing and theories that aid the design of both visual and auditory systems. The work process is described in section 4 along with explanations of the chosen methods. The results from the methods are presented in section 5. Both sections are divided into subsections named in the same way as the project phases. Section 6 discusses the results and answers the research questions.

2 Context

A transmission in a vehicle is used to change the speed and is installed between the engine and the driving wheels. Its main functions are to enable the vehicle to move from rest, adapt power flow, and enable reverse motion, locking engine power transmission and control power matching (Lechner, Naunheimer & Ryborz, 1999). The desired gear ratio is most often selected by moving a lever or knob, which is usually located between the driver and passenger seats.

2.1 FROM MANUAL TRANSMISSION TO SHIFT-BY-WIRE

Manual transmission (MT) shifters allow the driver to select a gear ratio (three or more forward speeds and reverse) while driving. This is done by navigating a shifting pattern to choose a desired gear.

Automatic transmissions (AT) also use gear levers but instead of the driver having to choose a gear, the AT automatically changes the ratio between the engine and wheels of the vehicle. Contrary to MTs, the gears aren't shown as numbers or chosen in the same way. ATs use positions that are denoted as drive ("D"), reverse ("R"), neutral ("N") and park ("P"). Cruiser or manual options can also be available. Just as in the case with MTs, the different gears are chosen by moving the shifter along a shifting pattern. When the transmission is in "D", the driver has to press down the accelerator pedal and the transmission will shift automatically through its forward range of gears. As the car loses speed the transmission shifts to a lower gear (Britannica Academic, n.d.).

A comparison of the physiological activity of the drivers using ATs and MTs has shown that the arousal levels are higher when using a MTs (Zeier, 1979). Automatic transmissions most likely increase the traffic safety by reducing the distractions, off-road glances and stress levels. On the other hand it can be a positive thing to engage more actively in the gear changing process during mundane driving situations. ATs are heavier than MTs and, with some exceptions, are less efficient than MTs (Consumer Reports, 2015; Lachnitt, 2013).

Most ATs that have electronically controlled actuation systems offer SBW functionality. SBW combines the ease of use of ATs with the lower cost, weight and fuel consumption from MTs (Dittmer, 2001). Furthermore, SBW require less force to use (Kobiki, Inoue, Sekiguchi, Itoh & Kamio, 2004) thus eliminating the wear-and-tear that happens in the mechanics between the shifter and the transmission (Serra Kia, 2015, 11 March). The electromechanical shift systems also provide a link to driver assistance systems such as automatic parking assistance, lane control and possibly more (Singh & Stolk, 2001).

2.2 POLYSTABLE AND MONOSTABLE SHIFTERS

A polystable shifter design has several stable points, which means that the shifters stays in place after a gear has been selected. The SBW shifters have introduced a new shifter movement pattern with only one or two stable position. This is known as a monostable shifter design and the shifter springs back to a starting position after a gear selection has been made (Lohilahti, Grane and Fristrom, 2016).

2.3 LAWS AND LEGISLATIONS FOR AT SHIFT LEVER SEQUENCES

Vehicles are sold globally and therefore different legislations and standards must be followed for the export to be possible. According to the International Organization of Motor Vehicle Manufacturers (OICA) in 2016 most car were sold in Asia/Oceania/Middle East and America. Car manufacturers need to follow the rules and standards for the countries to which they wish to export.

The National Highway Traffic Safety Administration (NHTSA) is a U.S. organization established by the Highway Safety Act of 1970 to enforce vehicle performance standards and reduce deaths and injuries from vehicle accidents (Federal register, n.d.; National Highway Traffic Safety Administration, n.d.). Their standards were used as a guideline for the thesis as they regulate what can be exported to the U.S. market. NHTSA has developed and enforces Federal Motor Vehicle Standards (FMVSS) as U.S. federal regulations (National Highway Traffic Safety Administration, n.d.).

The AT shift lever sequence for the controls and displays must follow a specified shifting pattern and gear positions must be denoted as separate identifiers (“P”, “R”, “N” and “D”) (49 C.F.R. § 571.101, 1968). “N” must be located between “D” and “R”. In the cases where “P” is included (not as a separate button) the indicator must be located at the end (foremost to the left or up), adjacent to “R”. Even if the transmission shift position sequence doesn’t include “P”, the identification of the shift positions and their relation to each other, as well as the selected position, must be displayed whenever the ignition is in a position in which the engine can be operated. This information doesn’t have to be displayed when the ignition is in a position that only allows starting the car. All the identifiers except for “D” must be shown as the abbreviations. “D” can be replaced by another alphanumeric character or symbol chosen by the manufacturer (49 C.F.R. § 571.102, 1968).

All the information that must be displayed should be displayed in the view of the driver. If possible, redundant displays providing the same or all of the information can be provided.

2.4 SHIFTER FEEDBACK IN VEHICLES

Throughout the thesis the term feedback refers to the information that is provided to the driver regarding the gear selection. Haptic feedback describes the sense of touch and means that the person receives the feedback through mechanical contact (Dijkgraaf, 2016). This feedback can be provided by changes in shape, vibrations, change in position etc. Visual feedback is given through visual stimuli. Examples of visual feedback are text, images, symbols, colors, shapes and more. Visual feedback is mandatory for AT and is often displayed next to the shifter as well as on the instrument cluster. Auditory feedback describes sounds that are produced in response to an action (Dictionary of Engineering, 2016). Examples include music, "beeps", clicks, voice commands and more.

2.5 PROBLEMS WITH MONOSTABLE SBW

The usability of monostable shifters has been questioned and studied after accidents were reported. During 2016 the NHTSA released a report stating that the ZF E-Shift monostable shifter, used in Fiat Chrysler and other vehicles, was connected to 314 complaints, 121 crashes or fires and 30 injuries related to its poor usability (U.S. Department of Transportation, 2016). Since then a death was reported as well and the cars with the installed shifter were recalled (Jensen, 2016, 23 June). The NHTSA report stated that the problems arose when people wanted to choose "P" but selected "R" instead, without understanding that they had done so. This resulted in roll-away accidents because drivers could open the door and step out of the vehicle while "R" was selected. At the time when the report was presented the only warning that was provided was a chime sound and a message on the instrument cluster that could easily be missed.

In the example above the problem was the poor communication between the system (car) and person, making it unintuitive. There can be many reasons for why the communication failed, bad feedback being one of them. The monostable SBW are quiet because the mechanical sounds that are associated with shifting MTs or ATs are removed, meaning that the auditory feedback isn't present. Auditory feedback is only there if it is added intentionally. Although visual feedback is provided through a shifting pattern and on the instrument cluster, the driver can't see which position the shifter is in by looking at the physical shifter (i.e. not the indicators). Finally the haptic feedback might also suffer in SBW if the manufacturer's desire is to make the shifting process smooth. Monostable shifters don't allow the user to put down and feel what position they are in since the shifters fall back into the same midpoint after the gear selection is made.

3 Theoretical framework

The implementation of design theory facilitates the communication between a system and people. In order to understand how to make a vehicle safe and accessible it is important to understand how humans interpret information, perceive visual and auditory cues as well as what difficulties arise in driving situations.

The theoretical framework defines usability, explains the Information Processing model and how feedback is used to draw attention. The knowledge was translated into a feedback system through theories regarding both visual and auditory information in vehicles such as Gestalt laws, color theory, the different types of auditory signals and what needs to be considered when providing the drivers with safe information.

3.1 INDUSTRIAL DESIGN ENGINEERING

Industrial design engineering is a professional field that develops products and systems for an optimization of function and appearance (IDSA, n.d.). The appearance of a product contributes to its perceived qualities and value (Johannesson, Persson & Pettersson, 2004). Color, shape, product identity, sound, usability and ergonomics contribute to this. Apart from the visual appearance, industrial design also takes into consideration knowledge regarding technology, economy, perception and semantics (Johannesson et al., 2004).

The products are developed through analysis and synthesis of data taking into account the requirements from the clients, the market, the manufacturer and environment. Analysis refers to the study of existing systems, its behaviors and properties. The gathered knowledge guides the synthesis process, where new concepts and solutions are designed (Johannesson et al., 2004).

3.1.1 Usability

The term usability has been defined by the International Organization for Standardization (ISO) as "extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use." (ISO 9241-11:1998). Effectiveness describes how accurate users achieve specified goals while efficiency focuses on the resources needed to complete the goal. Finally user satisfaction is defined as the freedom from discomfort, and positive attitudes towards the product (ISO 9241-11:1998). The benefits of usable systems are improved productivity, avoidance of stress, increased accessibility and reduced risk of harm (ISO 9241-210:2010).

The ISO definition of the term usability has been criticized for not being broad enough (Chen, Germain & Rorissa, 2011) and for not including the necessity of measurements (Schackel & Richardson, 1991). The proposed broader definition by Chen et al. (2011) includes visible working functionality, resemblance to similar systems, alignment with the environment, accommodation of the users' cognitive capacity and needs and good learnability. The main difference between this definition and the ISO definition is that it not only focuses on what is expected of

usable systems but also how those requirements can be achieved. The definition in itself includes ten principles of usable design (Jordan, 1998). In order for a system or product to be usable one must study the dynamic interaction between the user, task, tool and environment (Schackel et al., 1991). Furthermore, Schackel argued that it is important to specify what good usability is for a system in quantifiable terms.

3.2 INFORMATION PROCESSING

The different psychological processes that are involved when interacting with systems can be analyzed using the Wickens and Hollands (2000) Human Information Processing (IP) model, see Figure 1. The model describes how the stimuli are perceived, processed and analyzed.

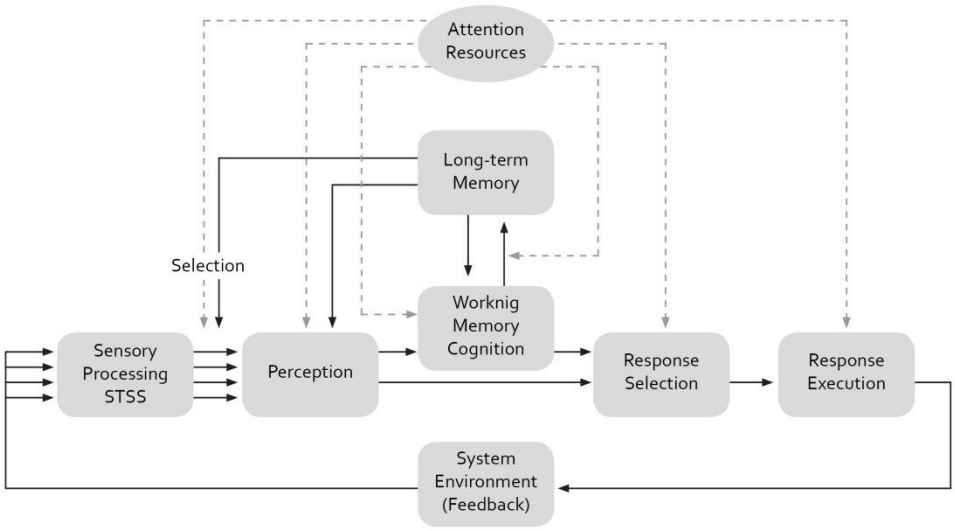


Figure 1. Wickens and Hollands IP-model.

According to the model the events in the environment are first processed by the five senses (sight, hearing, smell, taste and touch) and are briefly held in the short term sensory store (STSS). Because of the large magnitude of incoming senses, only a small portion of them are perceived (Wickens & Hollands, 2000). The STSS is divided into different types of memories depending on the type of stimuli. Iconic memory is the sensory memory for visual stimuli. It lasts between 1-5 tenths of a second, depending on the light and contrast. The sensory memory for auditory stimuli, echoic memory, is longer and lasts for 2-4 seconds (Danielsson, 2001). The meaning of the sensory signals is derived from past experiences that are stored in the long term memory (LTM) (Osvalder & Ulfvengern, 2008).

After the information has been processed it can either result in an immediate response or use the working memory (STM) to interpret the situation and choose a response. The STM has a limited capacity and can keep 5-9 units of unrelated information (Miller, 1956). The active information can easily be disturbed by new incoming information. Stress has shown to have an effect on the memory capacity and during some critical situations only one activity can be processed. Because of its small storage capacity there are risks for an overload of the STM (Osvalder et al., 2008).

The STM retrieves information form the LTM but it also creates a representation of the information in the LTM to be recalled at a later stage. The episodic declarative LTM memory is derived from personal experiences and is different from person to person. This is why it can be challenging to design visual or auditory icons that are recognizable in the same way for everyone, unless those are well known and standardized icons. The semantic memory includes knowledge regarding facts and words (Osvalder et al., 2008). When people hear the word "stop" most know what the word means, as long as they understand the language. By using speech cues the semantic memory is accessed and it is more likely that the cue is understood in the same way by different users.

The procedural non-declarative LTM memory is the how-to knowledge, such as walking, riding a bike etc. (Danielsson, 2001). Once a skill has been learned and stored in the procedural memory, it is highly unlikely that this will be forgotten. Those memories are usually acquired through repetition and practice. They are automatic behaviors but can get in the way when going from, for example, MTs to ATs.

Apart from the IP-model, the Situation Awareness (SA) model can also be considered when designing a communication system between a vehicle and user. It describes how the environment is perceived, understood and how this affects the near future (Endsley, 1988). In a driving situation this would mean how well the driver understands and interprets the information given by the vehicle system or the environment. If the information is interpreted incorrectly then the following actions would also be incorrect. This is what happened with the accidents related to the ZF E-Shift. The SA is affected by user expectations that are formed from previous experiences, communications or instructions. The expectations guide the attention and influence the perception of the incoming information (Endsley, 1994). This supports one of the design and usability principles where it is advised that new products are coherent with previous similar ones.

3.2.1 Multiple Resource Model

Mental workload can be compared to physical load where it becomes hard to hold on if the load is too heavy (Wickens et al., 2000). Referring to the IP-model, mental workload represents the proportion of available resources to meet a task. The available resources can be explained using the multiple resource model that includes four dimensions (stages, modalities, channels and codes) independent of each other, see Figure 2. Each dimension has a limited amount of resources within its levels. If two tasks demand the same level of a dimension, for example visual perception, then the two tasks will interfere with each other and less resources can be devoted to each task. If two tasks require different levels of a dimension, for example auditory and visual perception, then the interference is smaller (Wickens, 2002).

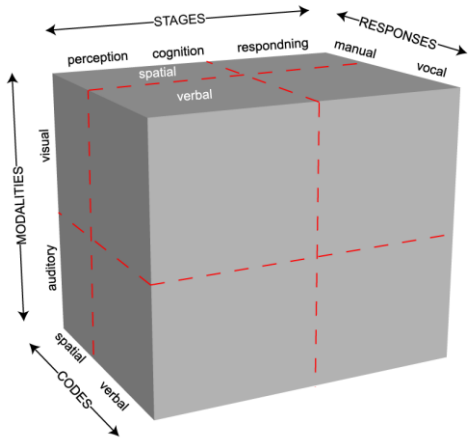


Figure 2. Wickens and Hollands multiple resource model.

3.2.2 Attention

Attention filters the sensory information and focuses the mental resources (Wickens et al., 2000). It can be distributed to several information sources, hence enable multi-tasking. However, since attention resources are limited, multi-tasking might fail if a certain task requires a lot of mental resources (Wickens et al., 2000). As the mental workload model has shown, it is easier to divide attention if the information channels are in different modalities (Bonnel & Hafter, 1998).

Attention is also an effect of the type and clarity of stimulus. Loud sounds, suddenly appearing bright lights, changes in contour, irregular movement in the peripheral visual field are some off the most effective stimuli that attract attention. The response to the sudden changes results in the fixation of the attention on the event that caused the response (Coren, Ward & Enns, 2004). Visual attention is drawn to objects that are large, bright, colorful and changing, for example blinking (Wickens et al., 2000). The peripheral vision is poor at picking up details but it is good at detecting changes in the visual field. This is especially so for sudden movements (Boyle, 2012).

People can direct their attention towards the objects of interest when searching for visual cues. Conjunction and feature searching are two different types of visual search, where both cases describe how the target object differs from its environment. Conjunction searching requires people to compare the target and distractors actively because of the relatively small difference between them (i.e. red dots among red triangles) (Coren et al., 2004). Feature searching results in identifying the target quicker because the difference between the target and distractors is greater in the sense that their features (orientation, color and/or curvature) are different. The number of distractors during feature change has almost no effect on the searching speed because the target simply “pops out” (Treisman, 1986). Complex or short representations of targets can cause errors (Prinzmetal, Henderson & Ivry, 1995).

Similar tasks should be grouped and organized so that the display is compatible with the task (Wickens et al., 2000). This can be achieved by implementing the Gestalt laws.

3.2.3 Feedback

Feedback means that when an action is carried out a response is given that acknowledges and confirms the action. The confirmation that is given through external sources, such as a computer or any technical system, can be visual, auditory or haptic. This feeding back of information is shown as a feedback loop in the IP-model (Figure 1) and it enables actions to become automatic over time (Boyle, 2012).

The learning process of a movement consists of three stages where the feedback is crucial. The first stage uses visual control (visual feedback loop), the second stage includes practice without or little visual control (proprioceptive feedback loop) and the final includes complete automation (no feedback loop) (Boyle, 2012). Complex practical skills such as driving rely on the automation of the action. That being said, automation can be hazardous because it is difficult to stop an automated action once it has been initiated. It also makes it difficult to go from one system to another, for example from MTs to ATs or polystable shifters to monostable (Boyle, 2012).

Correct feedback, provided during a movement, has shown to reduce errors in performance and minimizes errors in continuous tasks (Young, Schmidt & Lee, 2011). It can have a negative effect on the learning process though as people become dependent on the feedback in order to correctly perform a task (Schmidt & Wulf, 1997).

3.3 VISUAL INFORMATION FOR IN-VEHICLE DISPLAYS

Visual communication between a system and user is aided by Gestalt principles, colors, directions and symbols (Osvalder et al., 2008).

3.3.1 Gestalt laws

Gestalt laws imply that when presented with a number of stimuli people do not perceive each individual stimuli by itself, but instead perceive the objects in relation to each other (Wertheimer, 1938). The Gestalt laws are made up of several factors that contribute to why individual simple objects grouped together can be perceived as one complex object.

The Law of Proximity describes that objects located close to each other are perceived as one. Objects with similar properties, such as color or shape, are grouped by the Law of Similarity. The Law of Simplicity implies that the objects are seen in the simplest arrangement and form. The Law of Closure describes that objects close to each other are perceived as one, even if there are gaps between them. The brain simply closes those gaps, forming one complete object (Wertheimer, 1938). Objects close to each other and connected by straight or curved lines are seen as following a smooth path by the Law of Continuity (Cherry, 2016, 21 June).

3.3.2 Typography

Typography is the knowledge and technique of type, their arrangement and their connection to the environment. Depending on how typography is used, the designer can communicate different emotions, warnings or expectations (Bergström, 2012).

The fonts can be roughly divided into two families; roman types and sans-serifs, see Figure 3. Roman types include a serif, which is a slare or small line attached to the end of a stroke in a type. The serifs create a line in words and sentences, facilitating and leading the eye to read the text. Sans-serifs are types without serifs. Generally roman types are recommended for print while sans-serifs are recommended for screens. The low resolution of many screens make the serifs look like blurry points, counteracting their initial purpose (Bergström, 2012). In most cases the types for digital purposes should be simple, without serifs or italics (Simmonds & Galer, 1984).



Figure 3. Times New Roman (left) is a roman style. Arial (right) is a sans-serif.

There are many rules and definitions within typography that are used to design types in the best possible way, depending on the circumstances. Some of those are leading, kerning, legibility, readability and hierarchy. Leading is the vertical space between the lines of type, or text (Creative Bloq, 2016). Kerning is the horizontal distance between two letters. Too close or too far makes it hard to read or can give the impression that the letters don't belong to the same word. Legibility is how easily one letter is distinguished from another. It is closely related to readability, which is how easily text can be scanned by the eye. Hierarchy is used to separate the text and outline the most important information. It can be done with size (headings vs text body) or color, spacing and weight (Creative Bloq, 2016).

When deciding on the character size for in-vehicle displays, it is important to remember the vibrations caused by the driving task. The vibrations make it difficult to read small types. Too large characters, on the other hand, are difficult to read in one glance (Kimura, Marunaka & Sugiura, 1997).

3.3.3 Color theory

User experiences and population stereotypes shouldn't be disregarded (Wickens et al., 2000) when implementing colors in vehicle systems. The red color is most commonly used as a warning that needs immediate attention while yellow communicates a warning that isn't as urgent (Accurate Automotive Attention, n.d.). It could therefore be confusing if red is used to communicate selected gears.

The amount of colors used in a display should not exceed five or six colors (Carter & Cahill, 1979). Because the attention is drawn to colors, irrelevant color coding can be distracting. It is therefore important that different colors are compatible with the cognitive distinctions that are intended to be interpreted by the user (Wickens et al., 2000). Colors are seen differently with low illumination (night driving), for example red can easily be confused with brown (Stokes, Wickens, & Kite, 1990).

Everyone does not perceive color in the same way. 8% of the male population and 0.05% of the female population have color weaknesses (Colour Blindness

Awareness, n.d.). The most common blindness is between red and green, making it difficult to distinguish between the two (Boyle, 2012). Apart from color blindness, people can also perceive colors differently depending on where they live and their age. This is especially so for blue and green and has to do with the yellow pigment in the lens. The lens becomes more yellow with age and long-term exposure to UV-B or sunlight also leads to a premature raging of the lens. Because of this the blue color varies throughout the world, where it can be perceived more as green than blue in sunnier countries (Coren et al., 2004). Those facts mean that in design situations a combination of red and green as well as blue and green is advised to be avoided. Blue and green are also quite similar and might be confused in dim light situations where color vision becomes worse.

3.3.4 Contour and contrast

The contour is the fundamental building block for the visual patterns and can be defined as a sudden change of light intensity across space (Coren et al., 2004). Contours can be distinguished if there is a contrast in color or light intensity.

Apart from color and shape contrast, there is also a contrast in light intensity, which is the contrast between the brightness of a target and its background. It can be positive, negative or null (Noy, 2001). Furthermore positive contrast is bright objects on dark background and is preferred for vehicles in busy urban environments because it helps object recognition.

3.3.5 Consideration for visual information design

The ability of the eye to see details (visual acuity) differs depending on the environment. The acuity is better with higher levels of illumination while details are lost with poor illumination, which is one of the causes of night-time driving accidents (Coren et al., 2004).

Afterimages can be dangerous and are created when people suddenly see something that has a higher light intensity than its surroundings. As long as the afterimage is in the retina, that particular part of the retina can't see anything else (Boyle, 2012). Therefore white and strongly illuminated screens in vehicles should be avoided.

Astigmatism is a visual distortion that can be taken into consideration when designing information systems (Boyle, 2012). According to Boyle the visual impairment prevents focusing and produces blurred or distorted vision. The negative effects of the distortion become worse when high contrast is used; specifically white text on black backgrounds. If a dark background must be used, as it is often the case in vehicles, the problems can be avoided by using sans-serifs, thin fonts, allowing a distance between the letters or symbols and avoiding white objects (Anthony, 2011; Ali, n.d.).

The lens shape of the eye changes when people focus on objects at different distances, for example the signs on the road and the gear information on the instrument cluster (Boyle, 2012). This is known as the accommodation effect and it requires time, during which people are without clear vision. According to Boyle the accommodation delays are fractions of seconds but worsens with age.

3.4 AUDITORY INFORMATION FOR IN-VEHICLE SYSTEMS

Designing auditory cues takes the human physiology and perception into consideration. The sounds must be heard, not cause annoyance and contribute to an overall improvement of a product.

3.4.1 Hearable span and masking

The hearable span varies with age and physiology. A young, healthy person can hear frequencies in the span 20-20 000 Hz and sound levels 0-130dB (Bodén, Carlsson, Glav, Wallin & Åbom, 2001). As people age, their ability to hear high frequency sounds degrades. Speech varies in both strength and frequencies. Vowels typically have a low frequency and are easier to hear than consonants which have a high frequency and a weak sound (Bodén et al, 2001).

One of the main things to take into account, when designing audible warnings, is the background noise in which they will be heard (Edworthy & Heiller, 2001). Auditory masking occurs either when a new sound is presented to an existing one or when sounds are presented at the same time. Masking means that the person is unable to hear a sound due to another sound. The risks for masking are highest when the sounds have similar frequencies or if the target has a higher frequency than the masker (Coren et al., 2004). In order to design sounds that will not be masked by the environmental sounds, Patterson (1982) has proposed to predict the masked threshold and then design sounds 15-25dB above it. Speech is not as affected by the masking effect as other auditory cues and is easily understood even if the noise level is only 6dB less than the speech intensity (Coren et al., 2004). If the speech and background noise are the same intensity, people can still identify around 50% of the words.

Exposure to noise has both psychological and physiological effects on humans. It can worsen the attention, cause annoyance and decrease the task performance (Bodén et al., 2001). Furthermore, Bodén et al. suggest that sudden loud sounds can cause muscle tensions, increase in heart rate and blood pressure thus increasing the stress levels.

3.4.2 Types of auditory signals

Earcons are short and unique musical sounds that convey information about events (Blattner, Sumikawa & Greenberg, 1989). They transfer information instantly and are not largely affected by the masking effect in environments with a lot of speech (Parker, Eberle, Martin & McAnally, 2008). Earcons are not intuitive and must therefore be learned (Campbell, Richman, Carney & Lee, 2004).

Auditory icons imitate the sound they represent. An example of this is deleting a file on a computer that produces a sound that reminds of an object dropping into a trash container (Brazil & Fernström, 2011). They rely on social conventions for what different sounds mean and can therefore differ from location to location. Auditory icons should be used together with visual cues in vehicles and should not be used by themselves (Campbell et al., 2004).

Speech cues have the benefits of attracting attention without having to provide visual aid and require less learning than earcons and auditory icons. However, speech tends to annoy the users if it is presented too frequently. Speech should

only be used when the visual modality is overloaded in order to avoid problems with acceptance (Campbell et al., 2004). The speech should be kept simple, to a word or short phrase, for warnings or when time is critical (Campbell et al., 2004). Baber (1991) suggests that if speech is used, sound should differ from the human speech. This increases the awareness that the message is from the machine and not a human.

3.4.3 Considerations for auditory information design

Auditory information must be heard in order for it to have an effect. But it should not be too loud, cause annoyance or direct the attention from the primary task (driving). If the sounds are perceived as irritating, it can cause the person to simply turn off the warning or avoid purchasing a car with that particular auditory system.

Non-speech warnings need to be learned and are limited by the amount of different signals that can be remembered. Humans don't have problems hearing the difference between up to seven tone-based auditory warnings, but have problems hearing the difference when the number is exceeded (Stanton, 1994).

Regardless of what type of auditory cues is used, it is recommended that auditory information should be kept to a minimum (Edworthy et al., 2001). Too many warnings can accidently sound simultaneously and mask each other. The loudness can also be increased, making the driving environment unbearable. In a driving scenario, warnings and informational sounds can distract from more important auditory cues.

The Gestalt laws can be applied for auditory signal design as well as for visual signals (Hereford et al., 1994). The auditory cues are organized by rhythm, pitch, melodies or timbre. According to Hereford et al., the difference in rhythm is easier to tell apart than variations in pitch or timbre. If tones are used then they should not be played faster than four tones per second in order to be perceived.

3.5 DESIGN FOR TRAFFIC SAFETY

Vision is the main source of information when operating a vehicle. The driver is required to have divided attention and react quickly. The detected information is processed automatically with little awareness or effort. It has been suggested that one to two seconds can be used as a guideline for how long the driver can look at in-vehicle displays and away from the road (Zwahlen, Adams & DeBals, 1988). The risk of car accidents is largely increased if the off-road glance exceeds this number (Klauer, Guo, Sudweeks & Dingus, 2010), therefore it is advisable to design displays that require one to one and a half seconds off-road glances.

Sound is ephemeral while visual messages are permanent. However, visual cues demand that the driver looks directly at them to perceive the information while sound is not dependent on the direction in the same way. For this reason, sound should be used when immediate action must be taken and visual information for when the message is long (Hereford et al., 1994). A visual representation of the auditory information should also be available for future reference (Wierwille, 1993). It has been proposed that visual information is processed quickly and results in shorter response time than auditory cues (Lundkvist & Nykänen, 2016).

General ergonomic criteria must be met for in-vehicle displays. There must be good information visibility regardless of the road and the driver's sensorial capacities. The information must be quickly understood and not require more than two consecutive glances. The information must be immediately understood independently of the driver's cultural background or level. Lastly there should be no competition between sources of information (Labiale, 1997),

Stokes et al., 1990 has shown that pilots prefer visual over auditory warnings in situations that aren't urgent. This might also be applicable for drivers that might find auditory cues annoying. Experienced and novice drivers might require different warning and amount of information. A novice can require both visual and auditory cues, while an expert driver might want to turn off the auditory warnings. An option of tuning the warnings can be considered, but there is a risk that people overestimate their reaction time and choose to turn off warnings that would be helpful to them (Dingus, Jahns, Horowitz & Knipling, 1998). Regardless of the experience level, redundancy is recommended to be used in situations where there is a high risk of failure (Dingus et al., 1998).

4 Method and implementation

The thesis project was divided into six phases, see Figure 4. They include the common denominators in design work, which are to have a clearly defined aim, understand the problem and be ready for numerous iterations.

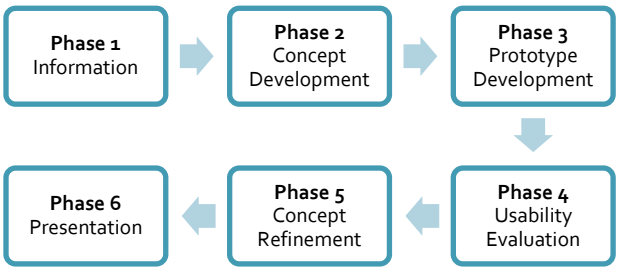


Figure 4. The six project phases.

4.1 PROCESS

The development process aims to improve existing solutions or present new solutions to existing needs (Karlsson et al., 2008). This particular project strived to improve an existing product by focusing on its usability and safety, therefore a human-centered approach was implemented. Human-centered design (HCD) is used to develop interactive systems that are usable by focusing on the users, and designing around their needs by applying knowledge in human factors and usability (ISO 9241-210:2010).

A cyclic development process was used in order to constantly evaluate the progress and re-adjust if needed to. This process includes going through different stages in the project several times, focusing on different aspects or phases in each round. The phases that should be included are planning, analysis of the problem, relevant research, concept development and evaluation, see Figure 5. As the project evolves and new things are learned it might be possible to go back to the research phase or adjust the plan.

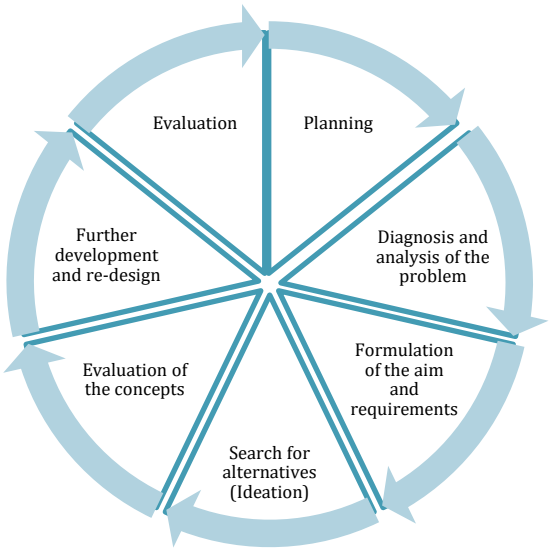


Figure 5. Project circle for a cyclic development process.

4.2 PROJECT PLANNING

Before the project was initiated, several steps had to be made to identify the problems and plan out the work. This meant understanding the problems associated with monostable shifters and preparing a project plan.

4.2.1 Defining the aim of the project

The project was initiated through a discussion with Camilla Grane and Sanna Lohilahti Bladfält, at Engineering Psychology at LTU, regarding their research that focused on the differences between monostable and polystable shifters. The conversation lead to the notion that the usability of monostable shifters might increase by providing the correct type of shifter feedback to the user, which ultimately became the main question of the project.

Before the project could be planned and defined, it was important to investigate whether or not it would be possible to create a functioning prototype that could be used for usability studies. The research department already had a physical shifter provided by Kongsberg Automotive and a laboratory set-up, but the shifter did not produce any electrical signals. Without being certain on how to make a working prototype, the project began with a planning phase that included prototype development from day one and a generous portion of hope and positivity.

4.2.2 Project phases

The project was divided into six different phases. They were reviewed and adjusted throughout the project, as expected in a cyclic development process. The six phases were: information phase, concept development, prototype development, usability evaluation, concept refinement and presentation. Several phases were dependent of each other, had to be done simultaneously and needed to be successful in order for the work to progress. The phases are presented below.

Phase 1 – Information. Review of the relevant literature and research, the problems with monostable shifters and a study of the existing SBW designs.

Phase 2 – Concept Development. First iteration of the development process. The requirement specification from the previous phase was used to design visual and auditory concepts, which had to be evaluated before making designs that would later be evaluated in a laboratory environment.

Phase 3 – Prototype Development. A working prototype had to be created to test the concepts from Phase 2. The prototype needed to provide direct feedback, which meant that both electronics and programming knowledge needed to be implemented.

Phase 4 – Usability Evaluation. Second iteration of the development process. The focus was placed on evaluating the designed concepts (in Phase 2) through usability tests using the prototype from Phase 3.

Phase 5 – Concept Refinement. Third iteration of the development process. The results from Phase 4 were used to design a final feedback concept.

Phase 6 – Presentation. The final phase is the documentation and presentation of the project. This included a master thesis and two separate presentations of the work progress and results.

4.2.3 Project planning

The plan for the development process was presented in a written project plan, which served as a foundation upon which the rest of the project was built. Project plans include information regarding the aim, time frame, planned methodology and anticipated project outcome (Johannesson et al., 2004).

A Gantt-chart are common project management and was used to plan the 20-week development project. It gave a quick visual overview over the different project stages, their dependencies and the allowed time (Wallace & Gantt, 1923). Figure 6 below shows a simplification of the complete Gantt-chart in Appendix 1, which was made with Microsoft Excel. The time frame is shown on the horizontal axis in weeks and the phases are placed vertically as blocks.

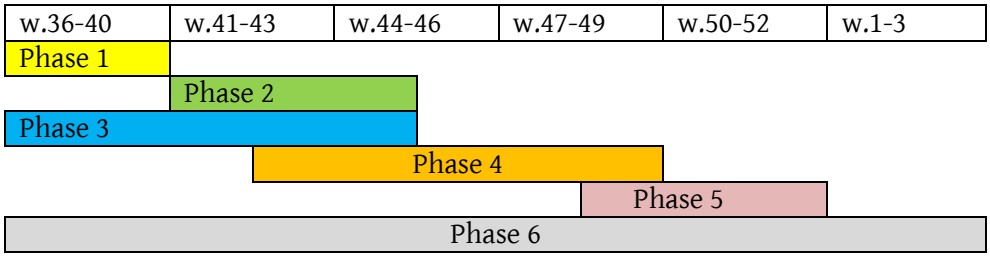


Figure 6. Simplified version of the Gantt-chart used for the project plan.

4.3 PHASE 1 - INFORMATION

Throughout the information gathering phase a version of the Needfinding process was used. The classic version aims to identify needs that people have but may not be aware of (Patnaik & Becker, 1999). The task was already given as well as the physical prototype, so the Needfinding focused on what information the drivers need in order to select the correct gear.

4.3.1 Literature review

The information presented in the theoretical framework was gathered primarily using Google Scholar (http://www.scholar.google.se) and the LTU digital library database (http://www.ltu.se/ltu/lib). Through the database access was granted to ISO standards (www.iso.org), engineering reports and publications by SAE (http://www.sae.org) and IEEE (http://www.ieee.org). Examples of search words were "shift by wire", "monostable shifter", "electric shifters", "feedback in vehicles" and more. Apart from engineering and scientific articles, printed literature was also used. The laws and regulations were found at the NHTSA website (http://www.nhtsa.gov) and the FMVSS were read digitally as there was no access to the physical regulation book.

Google search engine (http://www.google.com) was used to find articles about different shifter designs. The information from both Google and YouTube were compared to several resources before giving any credibility to the information.

4.3.2 Context immersion

Polystable and monostable shifters were observed, specifically the feedback that was given to the users by them. Observations are thought to be an objective method to gather information on how people behave. This method provides information on how the objects are truly used, which may not always align with how they are intended to be used (Kylén, 2004). The objectivity can be questioned, as what is observed can depend on the observer, expectations and the preexisting knowledge. The polystable shifters were observed by studying the behaviors of the drivers and taking notes of all the feedback, both intentional and unintentional, given by the shifter. The drivers were also asked how they used the feedback that had been observed. There was little access to monostable shifters, therefore they were observed by watching instructional videos on the websites of vehicle manufacturers and on YouTube (http://www.youtube.com).

An analysis of the existing and conceptual SBW designs was done through benchmarking. Benchmarking is used to analyze and evaluate competitor's products or solutions. This knowledge can then be implemented in the design process by avoiding the mistakes that other manufacturers make and including the positive aspects from their work (Johansson & Abrahamsson, 2008). The automotive benchmarking service A2Mac1 (http://www.a2mac1.com) was the primary resource for benchmarking concept gears from 2005 to 2014. The different design solutions were assessed and divided into two groups; one which included good designs and the other included bad designs. The definition of what was considered good or bad was based on the theories studied in the theoretical framework. The focus was placed on the usability of the designs and not on the appearance or innovation. The groups were analyzed to understand the common factors that contributed to good or poor usability.

4.3.3 Requirement specification

Requirement specifications establish the requirements that must be met for a design solution to be acceptable and they may not include any solutions within themselves (Karlsson et al., 2008). Two separate requirement specifications had to be written; one for a visual concept and one for an auditory concept. They were formulated using information from the theoretical framework and benchmarking.

4.4 PHASE 2 – CONCEPT DEVELOPMENT

Both auditory and visual concepts were designed and assessed in this phase. The concepts were constantly evaluated and critically analyzed, ensuring that preference would be given to concepts expected to be more usable rather than appealing.

4.4.1 Visual concept development

The creative process began by focusing on the overall visual concept design, without going into details regarding the colors. The overall design concepts were created using the Brainwriting method using pens and paper. Just as with brainstorming the aim is to generate as many ideas as possible. The difference between the two is that the ideas were sketched or written instead of spoken (Osvalder, Rose & Karlsson, 2008; Innovation Management, n.d.). After a decision was made on the direction that would be taken, the design process began narrowing down on details such as the font or specific color combinations.

A version of the Incremental UX method was implemented to design the details of the visual concept. The original method is used for interactive systems and aims to have a planned progression of features that add functionality to the design (Ritmeyer, 2015, 29 September). The same idea was practiced in the project, focusing on separate design elements at a time, using Adobe Illustrator. The first increment was targeted at the abbreviations. A shifting pattern, font, leading and kerning had to be decided upon. The second increment handled colors for the inactive gears as well as the selected gears. The last increment studied different shapes that could be added to the selected gear for reinforcement and redundancy.

An additional visual concept was worked on parallel to the incremental one. This concept was intended to be a Dark Horse, a concept significantly different from the others. It forces the designers to use a fresh approach, introduces “impossible” ideas and encourages innovation (Bushnell, Steber, Matta, Cutkosky & Leifer, 2013). These concepts were also first sketched out using Brainwriting and then refined in Adobe Illustrator.

4.4.2 Auditory concept development

Earcons, created in Proppellerhead Reason, and human as well as machine voices were evaluated. The human voices were recorded from two different people, one man and one woman, using the Apple Garage Band software. Machine speech was generated by text to speech websites. Acapela-group (http://www.acapela-group.com) was used to generate Swedish speech and Ivona (http://www.ivona.com) was used for the English voices. The machine voices included male and female voices. Two different accents were tested for the English speech, American and British. The voices were edited using Audacity (http://www.audacityteam.org). Both complete words (i.e. “park”, “reverse” etc.)

and abbreviations (i.e. “P”, “R” etc.) were generated. Voices that made it hard to hear the separate abbreviations were excluded.

All earcons and speech signals were tested with a driving white noise sound, created by Soundbuds™ on YouTube, to control if a masking effect occurred. This was tested for situations with no rain (Soundbuds™, 2015) and heavy rain (Soundbuds™, 2015).

4.4.3 Workshop for evaluating the concepts

A workshop was held to test the design concepts on a small group of people and to detect possible usability issues. During the early iteration in a design process five people is sufficient to discover major issues or preferences (Molich & Dumas, 2008). Five people participated in the workshop, three women and two men.

The workshop studied both visual and auditory concepts, starting with the visual concepts. The aim of the visual workshop was to learn about attitudes toward colors, shape and to see if anything form the dark horse could be implemented in the final concept designs. The auditory concepts were evaluated to learn about attitudes towards auditory cues, language and gender.

Information about the participant’s attitudes was quickly gathered through a Post-Up method (Straker, 1997). People were asked to write one word per sticky note and then discuss and group similar words on an A3-sheet that was prepared in advance. Opinions and collective attitudes regarding the auditory concepts were collected by Dot voting (Michael, 2011, 25 August). The participants were asked to vote on the preferred gender or language by placing a sticky dot next to their choice.

4.4.3.1 Procedure for visual concepts

The chosen color concepts were shown to the participants. They were asked to write the first three words they thought of when seeing the different colors on sticky notes, one thought per note and three notes per color. No discussions were allowed to avoid influences. After all colors had been shown, the group received four A3-sheets, one for each color (Figure 7). The participants were asked to post their notes, grouping similar words together. During this stage the participants were encouraged to discuss their choices with each other and explain their reasoning.



Figure 7. Grouping of Post-It notes on an A3-sheet in the workshop.

The different shapes that were added to the active gear selection were evaluated in the same way as the colors, but the Post-Up method was not used. Instead the group was presented with a separate A3-sheet for each concept. The sheet included a horizontal arrow that went from "bad" to "good". The group was asked to paste their sticky notes for the concepts along the arrow (Figure 8). They were also encouraged to discuss during this process. The Dark Horse was evaluated using the same method as the colors.

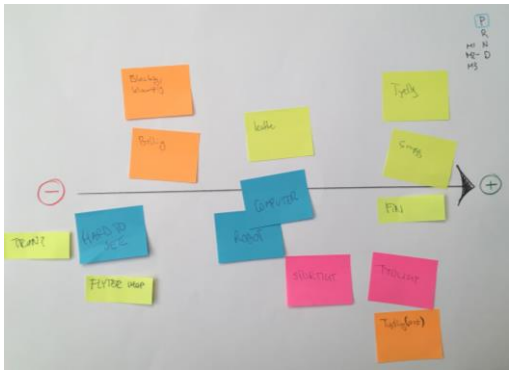


Figure 8. Evaluation of a visual concept in the workshop.

After all the visual concepts had been tested, each participant was given a ranking sheet. They were asked to rank the concepts with a number 1-5, where 5 was good and 1 was bad, and discuss their choices.

4.4.3.2 Procedure for auditory concepts

The auditory concepts were played to the participants, demonstrating how it would sound if they were to go from "P" to "M3". After each concept was demonstrated, people were asked to write down the first three words they thought of after hearing the voice. One though per sticky note. The order in which the voices were played was: Swedish male, American male, Swedish female and American female. The sticky notes were used in a Post-Up method and additional dot voting was implemented to analyze the language and gender preferences.

The workshop also aimed to examine how well the participants heard the concepts when other sounds were present. The participants were asked to sit in front of speakers that played a white noise simulating a driving situation (Soundbuds™, 2015). A smartphone was placed with a podcast from P3, a Swedish radio station. The "Energidryck vs kaffe, haj-update och bråkpasset i P3" episode in the "Morgonpasset i P3" pod-series was selected (Sveriges Radio, 2016, 18 October). It included male and female voices as well as various melodic sounds. Having this noise, the sequences were played once gain in the order: Swedish male, Swedish female, American male and American female. The participants were asked to rate how well they heard the speech cues and how well they recognized what was being said. The rating was 1-4, where 4 was best and 1 worst.

4.4.4 Concept selection for the usability evaluation

The information from the workshop was used to develop concepts for a usability evaluation. Apart from the analysis of the workshop data, the final concepts were also controlled using the requirement specification and the theory.

One of the research questions in the thesis was to study if auditory, visual or multimodal feedback was best in terms of usability. Therefore it was critical that the concepts were comparable, meaning that they had to be as similar as possible except for the factors that were being assessed (i.e. the modality). However, visual feedback tends to be permanent while auditory is temporary. For an auditory and visual concepts to be comparable, they must be displayed for an equal amount of time. This meant that either a temporary visual concept, that lasted for the same amount of time as the auditory, had to be included or a permanent auditory one.

4.5 PHASE 3 - PROTOTYPE DEVELOPMENT

A prototype was built and programmed for the usability evaluation of the chosen concepts. A mechanical monostable shifter prototype, provided by Kongsberg Automotive, was used as a base. To produce feedback signals a computer mouse that allowed programmable macros was needed for the shifter inputs. The mouse sensors were removed, adapted and fitted with flat springs to allowed inputs from the shifter. The springs were placed so that the motion of the shifter activated the correct sensors.

The input from the sensors emulated a keyboard. Four inputs (the sensors) were available but there had to be seven outputs (the gear selection). This problem was solved by using a flowchart that showed the decisions and their results, thus demonstrating the dependencies that would be used for programming. Flowcharts consist of defined various shapes and symbols, see Figure 9, to understand information processing problems and their solutions (ISO 5807:1985). An oval symbol means start or end, a parallelogram is an input or output and a rectangle is a process. The symbols are connected by arrows. ClickCharts Diagram Flowchart by NCH Software (http://www.nchsoftware.com) was used to create the flowchart.

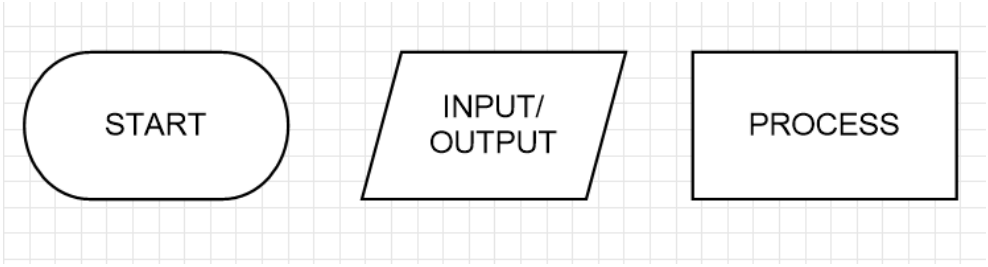


Figure 9. Flowchart symbols.

The dependencies were referred to when programming the feedback conditions in E-Prime. The software was developed by Psychology Software Tools Inc. (http://www.pstnet.com) and is often used in psychology experiments. The programming language is E-Basic, which was derived from Microsoft Visual Basic (http://www.microsoft.com). E-Prime made it possible to program the shifting pattern, feedback and instructions (see section 4.6). The programming depended on the concepts as well as the design of the usability tests.

4.6 PHASE 4 - USABILITY EVALUATION

The objective of the usability study was to analyze the effectiveness, efficiency and user satisfaction of the designed concepts. It also aimed to understand if visual, auditory or multimodal feedback resulted in the most usable feedback design.

4.6.1 Participants

The test participants (TP) were students and employees at LTU. 25 people participated with the age span 22-62, the average age was 29.24 and the median was 26 years. The test included 12 women and 13 men. Everyone except for one person had a driving license. The average driving experience was 9.8 years and the median was 7 years. The average driving frequency was "a couple of times a week". In average the participants were not experienced in driving with ATs, using it a couple of times per year. Eight TP had never used an AT prior to the usability test. The background information was gathered from survey (Appendix G) that was presented to each participant at the beginning of the test, right after they had signed a consent form (Appendix F).

4.6.2 Experiment design

The usability of the chosen concepts was evaluated in a laboratory environment through a usability test with a duration of approximately one hour. A within subjects design was implemented, allowing one TP to test all concepts. Apart from the fact that less people are required to participate, the error variance associated with individual differences is also reduced in within subjects design compared to between subjects designs (Psychology World, n.d.).

The order of the presented concepts was counterbalanced with a Latin Square. A Latin square is an orthogonal nxn mathematical model that contains n different symbols that appear once in each row and column (Denés & Keedwell, 1991; Williams, 1949). Since there were five concepts, the Latin square was of order five and had n=5. A simple algorithm was used to produce cyclic squares, which means that the elements in the columns were obtained by adding the same difference to the previous columns (Bradley, 1958). Table 1 explains the algorithm for a 5x5 Latin square. When the maximum value was achieved the next row starts over from the minimum value.

Table 1. An algorithm for a 5x5 cyclic Latin square.

1	2	n = 5	3	n-1 = 5-1 =4
2	3	1	4	5
3	4	2	5	1
4	5	3	1	2
5	1	4	2	3

A sequence of eleven shifting tasks was performed for each concept. A task denoted going from one gear position to another within a given 13s interval. The tasks had different levels of complexity depending on the amount of steps that the user needed to make to choose the correct gear. Five sequences with different tasks were prepared. A consistent level of complexity among the sequences was controlled by ensuring that the total sum of the steps used in the tasks was equivalent for all sequences (23+/-1) (Appendix D). It was also controlled that concepts did not always have the same sequences assigned to them (Appendix E).

4.6.3 Design concepts

The experiment included five feedback concepts all of which showed the shifting pattern on the instructional screen. The tested concepts were:

- Concept C – control concept with no dynamic feedback.
- Concept V – visual concept with permanent feedback that displayed the active gear through a blue background color (see Figure 10).
- Concept A – auditory concept where the active gear was conveyed by a female voice with a Swedish accent. The speech signal read "P", "R", "N", "D", "M1", "M2" and "M3" depending on the selection. The feedback was provided once with the selection and lasted for one second.
- Concept VT – identical to the visual concept with the exception that the feedback was displayed for a duration of one second. After one second the feedback disappeared and only the shifting pattern was shown until the next selection.
- Concept VA – a combination of concept V and A.

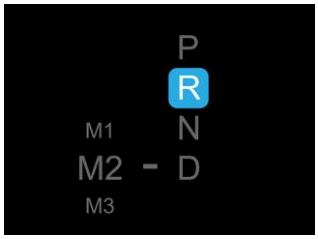


Figure 10. Visual feedback for concepts V, VT and VA.

4.6.4 Driving tasks

A Lane Change Test (LCT) driving simulator was chosen for the usability assessment. It is a three minute long standardized dual-task method used to estimate the demands of the secondary tasks on the driver, while simulating a driving task in a laboratory environment (ISO 26022:2010). A three lane road was displayed on a screen in front of the TP with signs that showed when a specific lane had to be chosen, see Figure 11. The image shows a situation where the right lane has to be selected. The lanes must be changed as quickly and controlled as possible using a steering wheel located in front of the driver.



Figure 11. Screenshot of a LCT-test.

4.6.5 Procedure

The test leader (TL) used a manuscript for the usability tests to ensure that all participants received the same information (Appendix B). Before initiating the usability tests, two training sessions were implemented. One was to practice driving with a LCT without a secondary task. The other training session was used to practice using the physical prototype without driving and feedback. The TL notified the TPs if an incorrect selection was made. Each TP had to make at least ten correct selections, three of which had to be consequent (Appendix C), before beginning the usability tests. The shifting pattern was displayed on an instruction screen to the left of the LCT screen (see Figure 12, bottom left corner).

After the training was completed the tests were initiated. Each concept was introduced and one selection was demonstrated. During the time the TPs navigated the LCT, a separate screen prompted them to selected new gears every 13 seconds. The instructions were given in a written and verbal form. The previously requested gear was the starting gear position except in the beginning of the test. For example if they were asked to select "R" in the previous task then the selected gear was "R" when the next task was given. The screen with the instructions was placed to left of the LCT screen, together with an image of the shifting pattern and the analyzed feedback. A camera placed in front of the TPs recorded if they looked away from the LCT screen to control the feedback or shifting pattern. The gears were selected using the produced prototype, which was located to the right of the TPs. Four cameras recorded the TPs; one in front, one recorded the shifter, one recorded the movement of the shifter and the last one recorded the LCT and instruction screen, see Figure 12.

After each concept was tested the TPs were interviewed and presented with surveys. Post-sessions surveys were handed out when all the concepts had been tested.

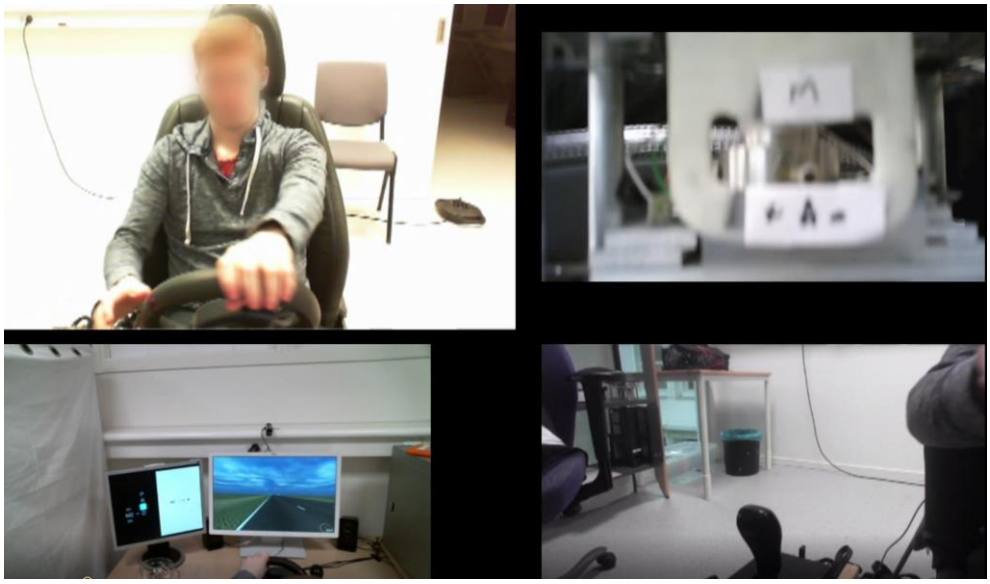


Figure 12. Recordings from a usability test.

4.6.6 Measurements
Different types of measurements were collected to control the effectiveness, efficiency and user satisfaction of the concepts.

4.6.6.1 Effectiveness
The effectiveness was measured through binary task success rates.

4.6.6.1.1 Task success
The task success rate had a binary definition (Tullis &Albert, 2013) where failure (0) was assigned when the TL had to notify the TPs that they had made an incorrect selection. Task success (1) was assigned when the TPs corrected their selection without any help from the TL

4.6.6.2 Efficiency
The efficiency was assessed by measuring the task completion time, off-road glances and the post-task NASA-TLX survey. The measurements were only taken for successful tasks.

4.6.6.2.1 Task completion time
The task completion time was measured with a stop watch on an iPhone using the recorded data from the tests. The unit was seconds with two decimals. The timing started after the instruction voice said "Välj" ("Choose") and stopped when the correct decision was made.

4.6.6.2.2 Off-road glances
Off-road glances were measured by marking how many times the participants looked away from the road after they were given a task until they chose the correct gear position. Observations were made to understand how the amount of glances off road varied for the different concepts. The recorded footage was used for the analysis.

4.6.6.2.3 NASA-TLX
NASA Task Load Index (TLX) is a workload assessment tool for subjective performance workload estimation of the users during human-machine interaction and interfaces. It is used to study visual and/or auditory displays and is gaining popularity for vehicle studies (Hart, 2006). NASA-TLX consist of two parts; a rating sheet that includes six factors and a weight sheet for selecting the factors that were considered to contribute most to the perceived workload (Hart & Staveland, 1988).

A paper version of the rating sheet was presented directly after each concept had been tested (Hart et al., 1988). Each factor had a scale that was divided into 20 equal intervals, anchored by opposite descriptors (high/low, perfect/failed). A translation made by Martin Englund (2014) was used for the factors, anchor points and explanations. The TPs rated their workload by a marking at the desired location of each factor (Appendix H). After all concepts had been tested, the TPs were asked to weigh the factors (see Appendix I).

4.6.6.3 User satisfaction
User satisfaction was collected by using post-task surveys (Acceptance scale, Product reaction cards) and short interviews, as well as post-session surveys (ranking sheet) and interviews.

4.6.6.3.1 Acceptance scale

A Van de Laan acceptance measured the acceptance of the feedback concepts (Van der Laan, Heino & De Waard, 1997). It is a simple and quick survey that consists of nine standard questions which are answered with a 5-point Likert scale, see Appendix H. A Swedish version, translated by András Varheli of LTH Lund, of the scale was used. The questions aimed to study the usefulness of the system and the user satisfaction. This method was designed specifically to measure driver acceptance of new in-vehicle systems. The questions were coded +2 to -2 from left to right, except for questions number 3, 6 and 8, which had the opposite coding. Questions number 1, 3, 5, 7 and 9 were part of the usefulness scale while questions 2, 4, 6 and 8 were used for the satisfaction scale. The usefulness scale was the sum of its items (1, 3, 5, 7 and 9) divided by five. The satisfaction scale was the sum of its items (2, 4, 6 and 8) divided by 4.

4.6.6.3.2 Product reaction cards

Product Reaction Cards (Benedek & Miner, 2002) captured the associations and attitudes TPs had towards the concepts. A list with adjectives was presented to the TPs (Appendix J). The participants were first asked to select all the words that they experienced described the concept. Once done, they were prompted to mark their top five selections. 89 adjectives were included in the list, which is slightly less than the original 118 words. Some of the words were excluded because they were not relevant. Each person received product reaction cards for two different concepts. In total all concept were evaluated ten times throughout the tests and every concept had 50 words associated to them. A matrix was used to control that the concepts had different orders when evaluated (i.e. concept "C" wasn't always evaluated when it was the first concept that was being tested), see Appendix E.

4.6.6.3.3 Ranking sheet

A ranking sheet was presented to each TP after all concepts had been tested and all other surveys were completed. The concepts were ranked with a number one to five, where five was given to the preferred concept and one to the least preferred (Appendix K). They were asked to motivate their answers verbally. The TPs were asked to motivate their choices.

4.6.6.3.4 Interviews

Unstructured interviews were held directly after the concepts were tested. The questions were:

1. What did you think of the concept?
2. Was there anything that you liked?
3. Was there anything that you disliked?
4. How did you use the provided feedback?

The interview form was relaxed and had a conversational tone. If a TP answered questions two, three while answering the first question then they were considered to be answered and not asked. Additional questions based on observations from the driving task could be included.

4.6.7 Data analysis

A short summary of what the TPs had said was written directly after each test session. They were read to analyze common attitudes among the TPs and interesting remarks that could contribute to the re-design.

The words from the product reaction cards for each concepts for all TPs were added and inserted into a Jason Davies (http://www.jasondavies.com/wordcloud) word cloud generator. The more common the words were, the larger the font was shown.

A repeated-measures ANOVA was performed on all the collected quantitative data (task completion time, off-road glances, NASA-TLX, Acceptance scale, ranking) using the IBM Statistical Package for the Social Sciences (SPSS) software (http://www-01.ibm.com/software/se/analytics/spss). A post-hoc Bonferroni corrected pairwise comparison was selected to investigate the differences between concepts.

Repeated-measures ANOVA can be used for within-subjects studies that have more than two sets of data, where the same subjects test different types of products. It analyses how the products relate to each other and if there is any difference between them rather than how people preform with the different products (Laerd Statistics, n.d.). The hypothesis of there being a difference between the tested variables is called the alternative hypothesis H_1. The alternative hypothesis is proved by disproving the null hypothesis H_0, which states that there is no difference between the variables (Gravetter & Wallnau, 2009). In order to know if there was a difference between the concepts or not the p-value for the within-subjects effects with sphericity assumed was controlled. The p-value is the probability that there is a difference between the concepts as opposed to the null hypothesis. The p-value must be less than 0.05 for evidence against the H_0. Generally, the lower the p-value the higher the probability that there is a significant difference between the tested subjects or concepts (Frost, 2014, April 17).

The results of the data analysis were presented in a defined manner which shows the F-value, the degrees of freedom, the error degrees of freedom and the p-value. All of the values are shown in the SPSS analysis as seen in the example in Figure 13. The highlighted numbers are the F-value (20.379), the effect of degrees of freedom (4), the error degrees of freedom (96) and the p-value (0.000). Those numbers should be presented as "F(4,96)=20.379, p=.00" (Van den Berg, 2014, 16 September). In the example the p-value is lower than 0.05, therefore the null hypothesis can be rejected.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
NASA	Sphericity Assumed	15473.200	4	3868.300	20.379	.000
	Greenhouse-Geisser	15473.200	2.798	5530.776	20.379	.000
	Huynh-Feldt	15473.200	3.204	4828.695	20.379	.000
	Lower-bound	15473.200	1.000	15473.200	20.379	.000
Error(NASA)	Sphericity Assumed	18222.723	96	189.820		
	Greenhouse-Geisser	18222.723	67.144	271.399		
	Huynh-Feldt	18222.723	76.906	236.947		
	Lower-bound	18222.723	24.000	759.280		

Figure 13. Example of Repeated-Measures ANOVA data.

4.7 PHASE 5 - CONCEPT REFINEMENT

The information gathered from the usability tests and interview insights were implemented to present a design suggestion for a feedback concept. The final concept was not further evaluated. The new design was illustrated using Adobe Illustrator.

4.8 PHASE 6 - PRESENTATION

The project was presented in a master thesis report and two oral presentations. The first presentation was held halfway through the work to update the examiner, supervisor and the rest of the thesis students of the work progress. The second presentation was held by the end of the project and focused on the work process as well as the results. In both cases each student was an opponent to another and commented both the report and the presentations.

4.9 METHOD DISCUSSION

The development process was cyclic and several iterations were made as new information was collected from the user observations, workshops and usability studies. Different sources of information ensured that the theoretical framework and benchmarking were reliable and had a scientific consensus. The concept generation process could have needed more than one workshop or an additional focus group study. Having a workshop provided information from users that weren't affected by the theoretical knowledge regarding the topic. Five people was enough to discover attitudes and issues with the concepts. However, involving people in the early stages of the development process, such as brainwriting, could have provided more innovative results.

The collected data from the usability studies provided objective information while the surveys, observations and interviews gave an insight into why a concept was usable or not. The sample size was large enough to counterbalance the concepts. The counterbalancing also ensured that concepts weren't always linked to the same sequence, thus avoiding the possibility of specific concepts having "easier" task sequences than other concepts. Including both men and women of various ages and driving experiences resulted in a heterogeneous sample group. That being said, the majority of the usability test participants were under the age of 30 and an unlikely target group for new cars. However, they will be the users of the

cars that will come in the future, therefore it might be a good idea to ensure that they understand the system.

Initially it was planned to include earcons, speech, icons and abbreviations. This was discarded because it would be difficult to compare earcons, which require learning, with visual abbreviations. The concepts had to be as similar as possible except for the studied variable in order to answer if a visual, auditory or multimodal concept resulted in better usability. Therefore if an earcon was to be used then a visual icon had to be used as well, which would go against the regulations and standards by NHTSA. To make the visual and auditory concepts comparable a temporary visual concept, with the same duration as the auditory concept, was designed (VT). The control concept with no feedback, concept C, was included to study if and how much different feedback concepts improved the usability.

Having a working prototype that provided immediate feedback was a key element of the project, without which the usability test would have been unreliable. The LCT test has its weaknesses, for example it isn't realistic that the vehicle would continue moving forward when "R" is chosen or that people would change gears while driving with an AT. Regardless of those factors, the LCT was chosen because the accidents that have happened with monostable shifters were due to distractions, stress or habits. Studying the feedback concepts while people were driving provided information on how well the concept perform when people can't only focus on the shifter.

The data collection, surveys and interviews were designed to gather a variety of different information about the efficiency and user satisfaction of the concepts. The task completion time and number of off-road glances provided information on how demanding the tasks were. The observations of the glances also revealed how people interact with the feedback. Both the Acceptance scale and NASA-TLX surveys were chosen to understand how satisfied with the concepts people were or how hard they perceived them to be. However, many people had a hard time understanding the Acceptance scale. The product reaction cards were effective but quite subjective, because people can attribute different meaning to the same words.

The analysis of the reaction time and amount of off-road glances had certain reliability issues because of the human error factor. E-Prime was supposed to calculate the time it took for the participants to make the correct choice for each task, but the coding did not work. Therefore everything had to be counted manually and the measured times can be questionable. However, the important factor was the difference in the time count between the concepts and not the actual time it took to make a correct selection. The time was always measured by the same person, therefore the error should be similar for the concepts. The same applied for the number of glances, which were measured by observation. It would have been better to use an eye tracking device because it would provide not only the amount of glances, but also the duration.

5 Results

The results for each phase are presented in a separate section, starting from the observations of existing feedbacks and ending with a final concept suggestion that resulted from the design process and usability evaluation.

5.1 PHASE 1 - INFORMATION

The information from the literature review is presented in sections 2 and 3. The following section presents the results from the benchmarking, the observations and the requirement specifications.

5.1.1 Observations of feedback in MTs and ATs

The driver received feedback from the behavior of the car when changing transmissions. MT shifters were navigated while the car was in motion. The automatic transmissions in AT shifters were used when the car was still, while the manual gears in ATs were used when the car was in motion.

Polystable MT shifters provided the driver with three different types of feedback. The driver received haptic feedback from the different stable positions. Without having to look down at the shifter position, the driver could feel where it was positioned and therefore the chosen gear. The different stable positions also gave the driver visual feedback. The positions were explained with a shift pattern usually located on the knob, see Figure 14 (Bobo_is_soft, 2006). Auditory feedback was provided from the mechanical sound that the change of gears made.

Polystable AT shifters provided the driver with haptic feedback in the same way as the MT shifters did. Apart from the visual feedback given by the positions of the shifter, the driver also received visual information about the chosen gear on the instrument cluster. In most cases the shift pattern was shown next to the knob (Figure 14, Upior polnocy, 1992), or on a small screen on the knob. The abbreviation of the chosen transmission usually had a contrast in light intensity or a color compared to the non-active transmissions. Similar auditory feedback to the MTs was given from the change of gears.



Figure 14. Left: manual shifter, Mazda Protege SE 1999. Right: automatic shifter, 1992 Ford Escort.

Regardless of the shifter the drivers were given different feedback, contributing to redundancy and strengthening the information. The observed drivers said that they used the visual information to control the shifting pattern or to confirm the chosen transmission. The visual information was rarely used by experienced drivers. The haptic feedback aided the drivers during the change in transmissions. The auditory feedback was not an intentional design but it also assisted in the transmission.

5.1.2 Analysis of the ZF monostable E-shift

The ZF E-shift was an electronic shifter that sprung back to a centered position after the driver selected a gear and released the shifter. The gears were chosen by moving the shifter different distances either forward or backward, depending on the desired selection. The monostable design provided some haptic feedback during the gear selection but did not provide any confirmation of the gear that was already chosen. Visual feedback was given through illuminated abbreviations on the shifter itself and on the instrument cluster, see Figure 15 (The Fast Lane Car, 2006, 22 June). There was no color or shape difference between the active and inactive gears. Because the shifter springs back, no quick visual feedback was given from the position of the knob. The instrument cluster showed the chosen gear in relation to the rest of the gears. The chosen gear had a higher light intensity but no color or shape contrast.

A chime sounded and a message was displayed when the driver attempted to open the driver's door while not in Park, see Figure 15. This has been taken care of since the NHTSA reports and after a software update Park is chosen automatically when the door opens.



Figure 15. ZF E-Shifter, Jeep Grand Cherokee (The Fast Lane Car, 2016, 22 June)

5.1.3 Benchmarking SBW feedback design

An analysis of eleven SBW designs (Appendix L) showed that in all cases visual feedback was displayed on instrument clusters and adjacent to the shifters. The information on the cluster showed the chosen gear. It was sometimes in isolation from the shifting pattern and sometimes together with the shifting gear, as in the case of Lincoln MKZ, Mercedes GLA45, 7G-FCT Mercedes Benz and some others. The visual information on the knob was shown as abbreviations along with the shifting patterns as regulated by FMVSS 101 and FMVSS 102. In most cases there was a color and illumination contrast between the active and non-active gears. In some cases there was also an addition of a shape, such as a dot, next to the active gear.

Auditory feedback was only used in two of the eleven designs that were analyzed. The Nissan Leaf and Toyota Prius had earcons assigned to "R". The produced sound reminded of the sound that trucks make during reverse. In the cases where manual gears were changed through separate paddles on the steering wheels (Mercedes GLA45 4Matic, Mesarati Quattroporte and Ghibli 2014), auditory feedback was given from the movement of the paddles.

The 2016 Acura TLX had a significantly different design where buttons were used instead of a knob (Appendix L). Visual information was provided by an illuminated colored contour around the active gear. Haptic feedback was given to the driver through different textures, shape and placement of the buttons.

The "P" gear selection was separated in BMW i3, Nissan Leaf and BMW 5 Steptronic by being activated through a button located on the knob. The pressing down of the button gave the driver haptic feedback. The separation of "P" from the rest of the gears decreased the risk of selection by mistake or missing the selection.

5.1.4 Good and bad visual feedback design

The theoretical framework, observations and benchmarking provided information of the aspects that contribute to good or bad feedback design. This only implies feedback from a usability standpoint and does not refer to the aesthetics or creativity of the feedback nor the physical shifter. The information below refers to the visual feedback shown on the knobs or next to the shifters.

Generally the visual feedbacks that were good from a usability standpoint had a strong light intensity and color contrast. The chosen gear was shown as both brighter and with a strong color, whereas the abbreviations of the gears that weren't chosen were in most cases white or grey. Rarely more than two colors were used, one for the active and one for the inactive gears. The most commonly used color was green but blue and red were also observed. The active gear was sometimes accompanied by an additional shape, for example a dot next to it. The Suzuki Concept Regina Gear Shift used redundancy as it showed the active gear both by a change of color and by a large letter on the knob, see Figure 16 (A2Mac1, 2011). The abbreviations used sans-serifs and were not too close to each other. In most cases relatively thin fonts were used.



Figure 16. Suzuki Concept Regina TO11 (A2Mac1, 2011)

The visual feedbacks that were not as usable had a little light intensity or color contrast between the active and inactive gears, as shown for the E-Shift in Figure 15. In some cases when an additional shape was added to the active gear, the abbreviation of the selected gear itself wasn't highlighted in any other way. In the case of the Tata Concept Pixel Gear Shift, left image in Figure 17 (A2Mac1, 2011), the line that leads to the active gear was illuminated but the abbreviation of the gear itself wasn't.

The visual feedback could be confusing when too many colors were used. The Kia Concept Soul Diva Gear Shift, right image in Figure 17 (A2Mac1, 2008), used four colors; white, orange, red and green. The position of the shifter revealed that "P" was chosen, along with the red dot next to it. Had it been a monostable shifter, the different colors would have contributed to a distraction.



Figure 17. Tata Concept Pixel GE1 and Kia Concept Soul Diva GE08 (A2Mac1, 2011 and 2008).

5.1.5 Requirement specification for the visual concepts

A requirement specification for the visual concept was written using the theories discussed in sections 2 and 3, as well as the analysis of the benchmarking results. All requirements had to be met in order for a concept to be approved and continue on to further evaluations.

The requirement specifications for the visual concept were:

1. It should be clear what gear is chosen
2. People should understand the feedback without having to learn in beforehand
3. The feedback should give provide information on how to select other gears from the chosen one
4. Abbreviations must be used for all gears. D may be an alphanumeric character
5. The used font must be easy to read during the day, night and by tired drivers
6. The order of the abbreviations must follow the regulations by FMVSS 102.

5.1.6 Requirement specification for the auditory concepts

Just as in the case with the visual concept, the requirement specifications for the auditory concept were derived from the theoretical basis presented in Sections 2 and 3, as well as analysis of the benchmarking.

1. It should be clear what gear is chosen
2. People should understand the feedback without having to learn in beforehand
3. The sound should not be masked by other noises that are normal in driving situations
4. The sounds may not remind of existing warning sounds in vehicles or sirens from service cars, i.e. police or ambulances

5.2 PHASE 2 – CONCEPT DEVELOPMENT

The concept development methods resulted in a visual concept that used a bright blue color for the selected gear, a shape difference between the active gear and the rest as well as a relatively simple design. People preferred to a female machine voice that pronounced abbreviations in Swedish.

5.2.1 Visual concepts

ATs usually have the shifting pattern adjacent to or on the knob and only show the active gear on the instrument cluster. Monostable physical shifters provide no quick visual feedback about the relation between the active gear and the rest of the gears since the spring back to the middle after each selection. Therefore, to satisfy the third requirement in the requirement specification mentioned in section 5.1.5, the visual feedback had to include the shifting pattern.

The shifting pattern and hence the gears were chosen to be shown as abbreviations with the placement according to FMVS 102. The visual feedback mimicked the physical shifting pattern, adding an element of coherence with both the mechanical movement and with other ATs. The Gestalt laws, specifically the law of proximity, were implemented to separate the automatic mode from the manual mode. The concept with a line between the "D" and "M2" was chosen because it indicated that the manual mode is accessed from "D". "M1" and "M3" were chosen as a smaller font to show that those were not accessible through "N". In final designs these would most likely be exchanged for "-" and "+", but were kept as abbreviations for the purpose of the usability study. Figure 18 shows the chosen design before colors were decided upon.

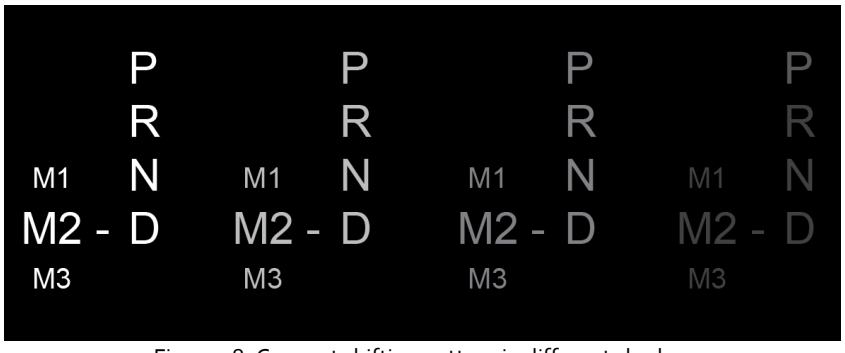


Figure 18. Concept shifting pattern in different shades.

A black background was chosen to work well during both day and night time. It was also chosen to blend in well with the rest of the in-vehicle systems. A light Arial was picked as the font for the abbreviations. It is a sans-serif and the capital letters, which wore used for the abbreviations, have a similar width. The similarity in the width made it look more cohesive and clean.

Only two colors were decided to be used in a concept; one for the active gear and one for the rest. The analyzed colors for the selected gear were blue (#29ABE2), green (#37B34A), yellow (#FFF100) and red (#EC1C24). The colors for the rest of the gears in the shift pattern were two different shades of cold grey, #A6A8AB and #6D6E70. These shades resulted in a light intensity contrast between the active and non-active gears because of the brightness difference. This also made the color "pop-out". Figure 19 shows the colors that were chosen.



Figure 19. Colors considered for the visual concept.

Different shapes were added to the active gear to separate them as much as possible from the rest of the gears, thus making them easy to quickly identify. The shapes that were chosen were a dot next to the active gear (concept DOT), a thin rectangle around the selected gear (concept BOX) and a color block inside which the active gear was be placed (concept BLOCK) (Figure 20). In the latter case three colors were used: gray, white and green or another color for the active gear.

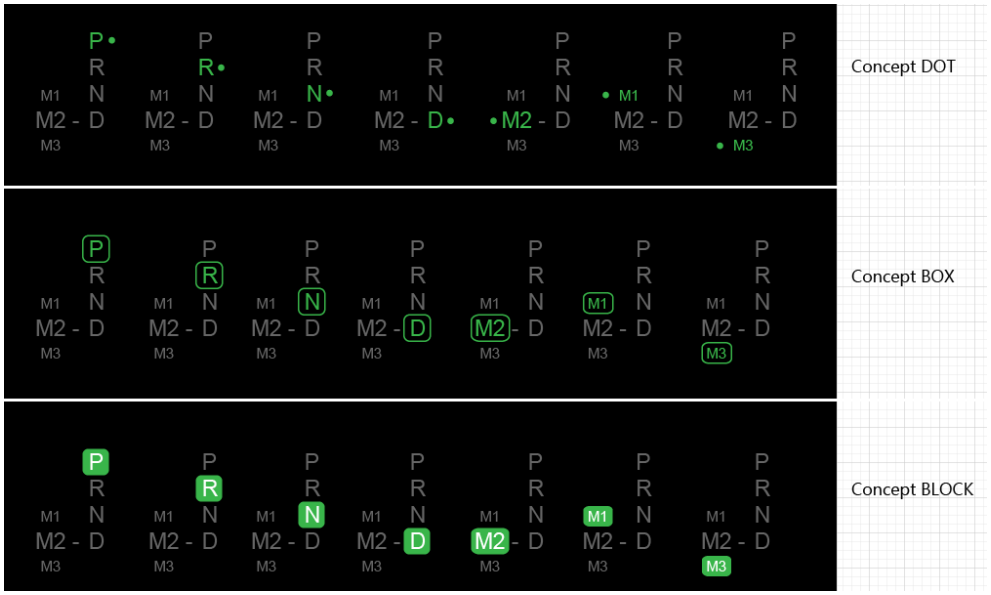


Figure 20. Shape variations for one of the visual concepts.

5.2.2 Visual Dark Horse

A visual dark horse concept aimed to be as different from the other visual concepts as possible, while still satisfying the requirement specification. The result was a concept that was called “Interactive”. All the visual concepts that had been designed included a static shifting pattern and a dynamic active gear. The active selection for the Dark Horse would always be in the same position and highlighted, but the inactive would move around as on a circular handle.

The manual and automatic modes were separated by belonging to two different blocks. The manual block was only highlighted if it was possible to access it through “D”. The automatic mode block was always highlighted since the automatic mode can be accessed regardless of the chosen gear in the manual mode. Figure 21 shows the Interactive design and how it behaved depending on the selection.

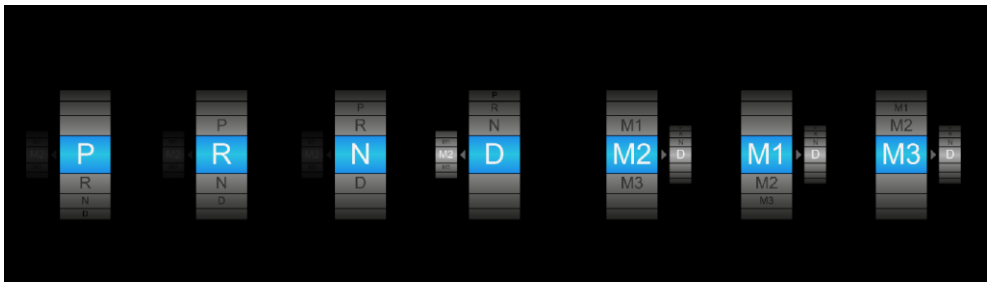


Figure 21. Visual dark horse concept.

5.2.3 Auditory concepts

Speech was chosen for the auditory feedback. The earcons demanded a learning process and would not have been comparable to the abbreviations in the visual concept. It was decided that there would be an auditory feedback for all the gears selected gears, once again for the visual and auditory concepts to be comparable. This mean that as people went from “P” to “M3”, they would hear the following: “R”, “N”, “D”, “M2”, “M3”. Abbreviations were chosen for the speech not only because of the coherency with the visual concept, but also because it was regarded as an annoyance factor if complete words would be said when going from “Park” to “Manual 3” (“Reverse”, “Neutral”, “Drive”, “Manual 2”, “Manual 3”).

Artificially sounding machine voices were chosen instead of recorded human speech to make a clear connection between the sound and vehicle. The American voices named “Salli” and “Eric” from Ivona were used because the letter “R” was easier to understand with an American accent than a British one. The Swedish voices “Emma” and “Emil” from Acapela were chosen as they had a neutral Swedish accent.

5.2.4 Workshop results

The participants associated the red color with a warning, stop or with an incorrect selection. The yellow color was interpreted as a warning and was confusing. The green color had the words "OK", kind, nature and environmentally friendly connected to it. The blue color generated the words "OK", safe, peaceful, clear and cold.

Concept DOT was unpopular, felt cheap and boring. Concept BOX was more popular but people expressed that it could be hard to see. Concept BLOCK had the most positive words associated to it and was thought of as clear. The Dark Horse was highly unpopular as it was thought to be too complex and that the design was too elaborate for a something that had to be simple.

The Swedish female voice ("Emma" from Acapela) was preferred. The women in the group said that they felt that the male voice was "mansplaining". The male voices were harder to hear when background noises were added.

5.2.5 Concepts for the usability evaluations

The concept development resulted in five different concepts, two of which were designed only for the usability tests and to answer the research questions. The selected concepts were a visual concept (Concept V), an auditory concept (Concept A), a multimodal concept (Concept VA), a control group concept (Concept C) and a temporary visual concept (Concept VT). Concept VT was identical to Concept V but the visual feedback disappeared after one second, lasting as long as the auditory information in Concept A. Concept C provided no feedback and was used as a control concept to investigate the effect of feedback. However, the shifting pattern (static visual information) was presented in all concepts, including Concept C. Concept VA was a combination of concepts V and A, with permanent visual feedback and temporary auditory feedback.

Concept blue BLOCK was chosen as the visual feedback in concepts V, VA and VT (see Figure 22). It was clear and was expected to be seen in the peripheral vision. The blue color was chosen because it was not used for warning in cars. Green was disregarded since it was associated with being environmentally friendly, which would not make sense if the people would drive with a gear that is above the recommended. The minimal design showed the shifting pattern, simplifying the gear selection with monostable shifters. The auditory feedback implemented in concepts A and VA was the female "Emma" voice from Acapela. It informed of the gear selection with abbreviation pronounced in Swedish. The abbreviations were announced as soon as the selection was made and only one time per selection.



Figure 22. Selection of the different gears for concept V and VA:

A clear distinction was made between the feedback cues and instruction cues. The feedback voice for concepts A and VA was female, therefore the male Swedish voice "Eric" was chosen to read the instructions in the usability study. The displayed written instructions were shown on a white background, contrasting the black used for the feedback.

5.3 PHASE 3 –PROTOTYPE DEVELOPMENT

A computer mouse that allowed programmable macros was used for the shifter inputs. The mouse sensors were adapted and fitted with flat springs to allow for an input. The springs were glued on a small box made out of rulers placed so that the motion of the shifter activated the correct sensors, see Figure 23.

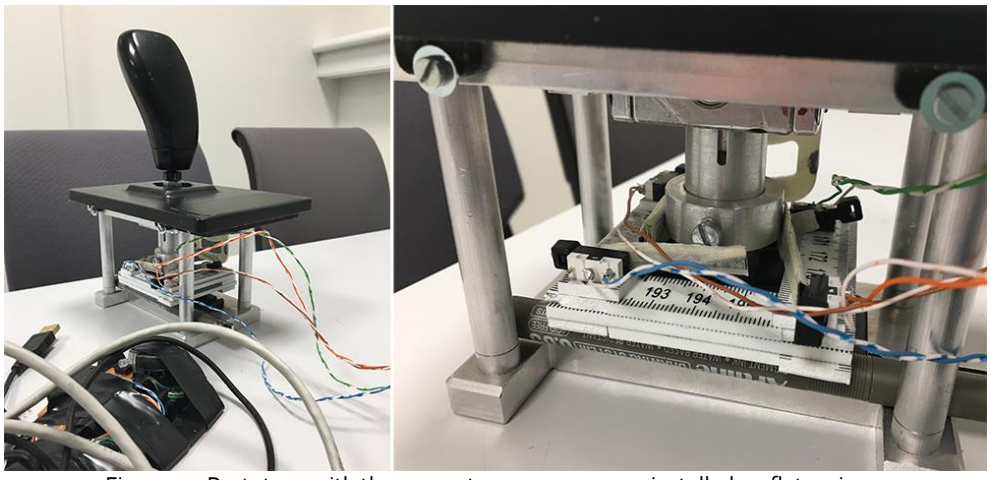


Figure 23. Prototype with the computer mouse sensors installed on flat springs.

The selected prototype had four directions in which it could go; forward, back, right and left. The forward and back motions caused the shifter to spring back to a middle position. The left and right motions locked the shifter in the position. The sensors that were placed were programmed to have the following input: forwards=f, back=b, automatic=m and manual=a. The shifter activated the sensors through its massive cylinder (depicted as red in the sketches below) that moved inside a box containing pathways adapted to the cylinder as shown in Figure 24. The figure demonstrates the motion of the prototype in automatic mode. The boxes under each shifter sketch is seen from above and shows the movement of the prototype's cylinder that activated the sensors through the pathways. The movement was similar in manual mode. The only difference was that in the front view, the red circle seen from above was placed to the left (a) instead of the right (m).

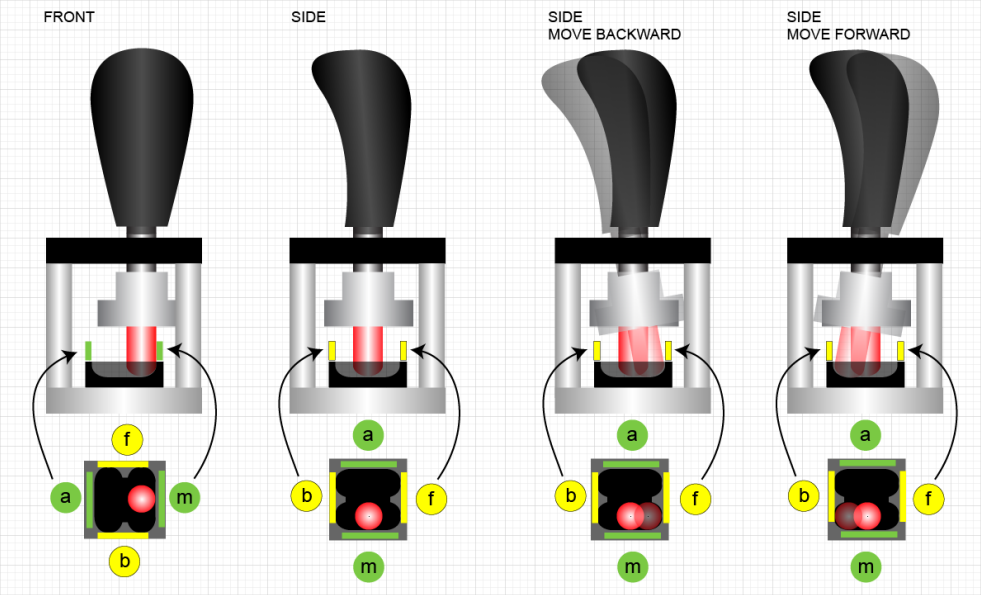


Figure 24. Sketch of the prototype motion in automatic mode.

The inputs from the four sensors were programmed to produce a feedback output on a screen, the conditions were sketched out in a flowchart shown in Appendix M. If a user was to go from "P" to "D", the shifter would have to be in the automatic mode (sensor m) and moved back 3 times (sensor b) through "R", "N" and finally "D". After each time the shifter would spring back to a middle position. The manual mode could only be reached from "D" and a warning message was given if TPs tried to access a manual mode from another gear in automatic mode.

5.4 PHASE 4 – USABILITY EVALUATION

A repeated-measure ANOVA was used to analyze the data for the task completion time, off-road glances, NASA-TLX and Acceptance scale. The measurements were assessed by answering the questions below:

1. Which concepts were significantly better than the control concept C?
2. What was the difference between the feedback concepts?
3. Was there any difference between concepts A and VT?

5.4.1 Task success rate

There was no difference in the task success rate between the concepts. The analysis revealed that mistakes were rarely made. In most cases the TPs could correct their own selection because they were notified by the provided feedback.

5.4.2 Task completion time

There was a significant difference between the feedback concepts ($F(4,96)=12.3$, $p=.000$), see Appendix O. Concepts V($p=.000$) and VA($p=.000$) were significantly better than Concept C. Concepts VT and A weren't better than C. Concept V was better than A($p=.000$) and VT($p=.016$). VA was also better than A($p=.004$) and VT($p=.030$). There was no difference between concepts V and VA. There was also no difference between concepts VT and A($p=1.00$). Figure 25 shows that there seem to be tendencies indicating that concept A took slightly longer time to navigate than C or VT.

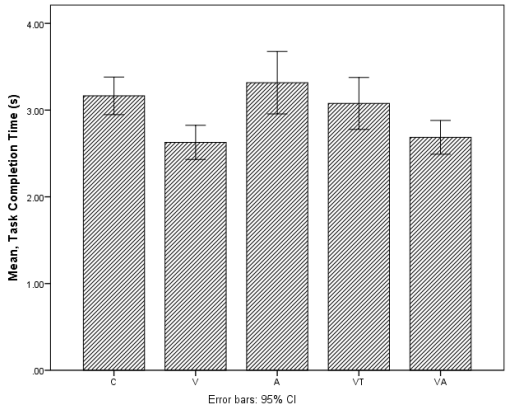


Figure 25. Task completion time in seconds.

5.4.3 Off-road glances

There was a significant difference between the feedback concepts ($F(4,96)=15.47$, $p=.000$), see Appendix P. Concepts V($p=.001$), A($p=0.001$) and VA($p=.000$) were significantly better than Concept C. Concept VT was not better than C. Concept V was better than VT($p=.035$) and worse than VA($p=.032$). Concept VA was additionally also better VT($p=.000$). Concept A was significantly better than VT($p=.002$). Figure 26 shows that auditory concepts required less off-road glances.

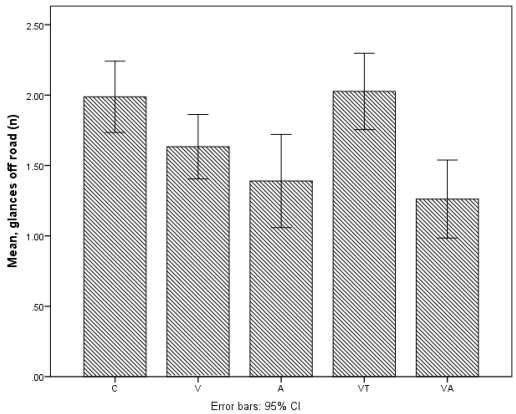


Figure 26. Amount of off-road glances.

5.4.4 NASA TLX

There was a significant difference between the feedback concepts (F(4,96)=20.38, p=.00), see Appendix Q. Concepts V(p=.000) and VA(p=.000) were perceived as less demanding than C. Concepts A and VT were not better than C. Concept V was better than A(p=.011) and VT(p=.007). Concept VA was better than A(p=.000) and VT(p=.001). There was no difference between concepts V and VA(p=.691). Figure 27 shows that there might be tendencies for VA to be less demanding than V. Concepts A and VT didn't have any significant difference(p=1.00).

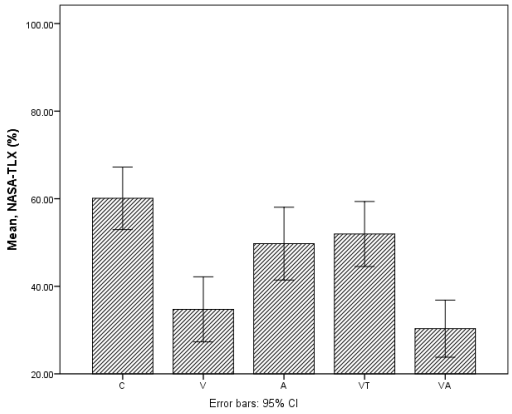


Figure 27. NASA-TLX results shown in percent.

5.4.5 Acceptance scale – usefulness

There was a significant difference between the feedback concepts (F(4,96)=20.38, (F(4,96)=17.82, p=.00), see Appendix R. Concept C was worse than V(p=.000), A(p=.018) and VA(p=.000). There was no significant difference between concepts C and VT (p=.076). Apart from concept C, V was only better than VT(p=.011). There was no difference between concepts V and VA(p=1.00). Concept VA however was better than C, A(p=.004) and VT(p=.003). As Figure 28 shows VA tended to be perceived as more useful than the rest of the concepts. There was no difference between concepts A and VT(p=1.00).

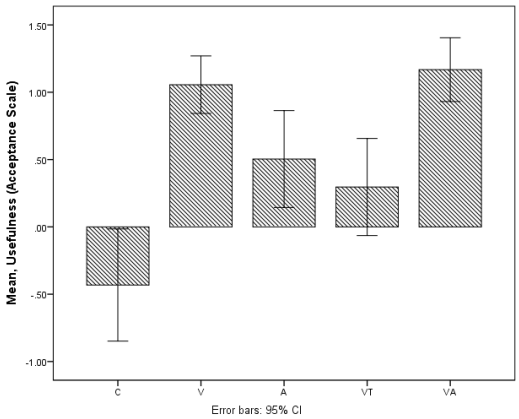


Figure 28. Usefulness results from the acceptance scale.

5.4.6 Acceptance scale – satisfaction

There was a significant difference between the feedback concepts ($F(4,96)=8.29$, $p=.00$), see Appendix S. Concept V was the only concept that was significantly better than C($p=.000$). V was also better than A($p=.004$) and VT($p=.014$). There was no significant difference between V and VA($p=.124$) but Figure 29 shows that there were clear tendencies for a higher satisfaction score for V. Concepts VT and A were perceived as equally dissatisfying($p=1.00$).

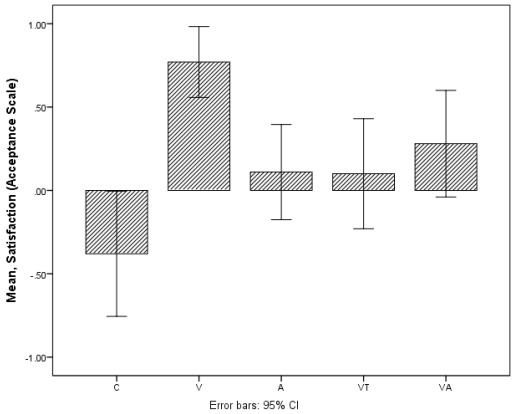


Figure 29. Satisfaction results from the acceptance scale.

5.4.7 Ranking

There was a significant difference between the feedback concepts ($F(4,96)=18.02$, $p=.00$), see Appendix T. Concept C was worse than V($p=.000$), A($p=.001$), VT($p=.000$) and VA($p=.000$), see Figure 30. Concept V was better than A($p=.005$) and VT($p=.001$). There was no difference between V and VA($p=1.00$). Concept VA was, apart from C, also better than A($p=.011$). There was no difference between concepts A and VT($p=1.00$).

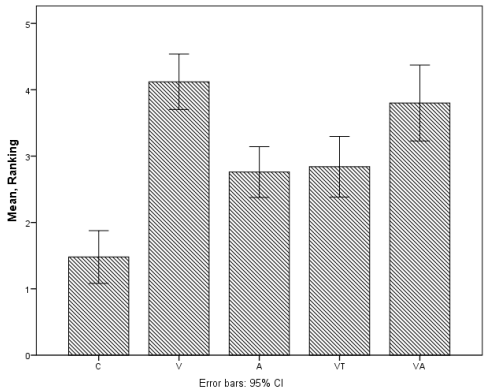


Figure 30. Ranking scores shown in ranking numbers.

5.4.8 Product Reaction Cards

The words generated from the product reaction cards are presented as word clouds in Appendix N. The most common words (selected three times or more) associated with each concept are displayed in Table 2.

Table 2. Results from the Product reaction cards

Concept	Most common words	Number of different words
C	Uncertain, unhelpful, simple, stressful, complicated, familiar	28
V	Understandable, easy to use, simplistic, familiar, expected, helpful, reliable	21
A	Helpful, understandable, annoying, reliable, friendly,	30
VT	Helpful, frustrating, unhelpful, stressful, distracting	32
VA	Reliable, easy to use, helpful, secure, disturbing, calm, simple, understandable	25

5.4.9 Interview results

Most feedback concepts were perceived as better than the control concept C. The concepts that were preferred by most participants were V and VA because of the permanent visual feedback.

The test participants reported that they could see the active gear in the peripheral view because of its position in relation to the shifting pattern. Even if they did not see the abbreviation inside the color block, the position, strong color and movement provided enough information. Based on the observations and user feedback, displaying the shifter pattern removed the necessity of remembering it. The interviews indicated that people either liked the sound and therefore preferred VA or highly disliked it and therefore preferred V. TPs perceived that the auditory feedback in VA helped them keep their focus on the road. The visual feedback was usually used to control the starting position or as a confirmation, while the auditory was used for navigation. Most participants reported that the sound was irritating when longer sequences had to be made and the abbreviations got interrupted. VA could be perceived as overwhelming and distracting.

VT was stressful for most people and in some cases worse than the control concept C. Even though VT was shown for the same amount of time as A, concept A was not perceived as stressful but it was seen as more irritating. Some reported though that they liked that VT only displayed the information when it was needed, thus it was seen as less distracting than V. In the cases when concept A was preferred to VT, people reported that the sound helped them focus on the driving task. According to several participants, the visual information was understood quicker than the auditory.

5.4.10 Summary

The data and interview analysis indicated that there was a clear improvement in the usability of the shifter when permanent visual feedback (concepts V and VA) was included. There was no significant difference in the data between concepts V and VA, but the interviews revealed that the participants were divided into two groups; one that loved the auditory feedback and one that despised it.

The temporary feedbacks (concept A and VT) performed equally poorly implicating that there might be no difference between the two modalities. Although it was observed that people became more distracted and had a harder time correcting their positions for auditory concepts if an incorrect gear selection was made. People could imagine having auditory feedback in certain situations when they do not wish to look away from the road. They requested that there should be an option to turn off the voice. They also mentioned that there shouldn't be feedback for each step when longer sequences are made (i.e. from P to D) as it was irritating and distracting.

The visual elements of concept V and VA were well received and could be seen in the peripheral vision by many of the TPs. They appreciated that the shifting patterns was always shown. It helped them see the relation between the chosen and requested gears, thus planning out the shifter movements.

5.5 PHASE 5 – CONCEPT REFINEMENT

The final concept was based on the results indicating that concepts V or VA had the best feedback from a usability perspective. Since the auditory information was disliked by some participants, the final concept VRA (visual and reduced auditory) included permanent visual feedback and optional auditory information.

The visual feedback, displayed on the instrument cluster, consisted of a shifting pattern with a blue color block on the chosen gear (Figure 31). The same feedback was displayed on the physical shifter. This made the chosen gear easily identifiable and seen in the peripheral view. Arial font was kept for the gear abbreviations. A line connected "M" to "D" and the signs "+" and "-" were used for the manual gear selection. In cases when the manual mode was selected, the changes were shown in the "M" abbreviation with numbers depending on the gear selection (see Figure 32).

An option to activate or deactivate the auditory feedback through a selection on the shifter was included. A sound icon was displayed on the instrument cluster when the auditory feedback was selected (Figure 31) and the same icon was darkened when the sound was deactivated (Figure 32). If auditory feedback was activated, it was only given for the gears where there have been documented problems with monostable shifters. Those were "P", "R" and "D". The sound was a female machine speech signal with a Swedish pronunciation of the abbreviations.



Figure 31. Concept VRA with activated auditory feedback and selected "D" gear



Figure 32. Concept VRA in manual mode with deactivated auditory feedback

6 Discussion

Results from the usability studies and user feedback have shown that the usability of monostable shifters could be improved through feedback design. Permanent visual feedback must be included while auditory feedback could be implemented if it is seemed necessary. In cases when auditory information is included it should be used sparingly and together with visual information.

6.1 RESULTS

The design development process and analysis resulted in a multimodal VRA concept that included permanent visual feedback and optional auditory information. The redundancy could help people who are not used to monostable shifters learn the sequence and eventually automating the movement. With growing experience they might want to turn off the sound. They could also choose to turn on the sound in situations where a large amount of resources from the visual modality are needed. In these cases the auditory information could help avoid overloading the visual modality and unnecessary mistakes.

VRA used the same type of visual feedback as concept V and VA. It was easily understood as it looked familiar to the shifting pattern usually displayed adjacent to knobs in ATs. The visual separation between the automatic mode ("P", "R", "N" and "D") and the manual mode indicated, through the Gestalt laws, that the two modes were different. The usability test participants had no confusions regarding what the visual feedback meant. One reason could be the simplicity of the test and that this was the only thing, except for driving, that they had to focus on. Even if this was the main reason, the design factor should not be disregarded. Only three colors were used to keep the design simple and avoid uncertainties. The color, shape and intensity contrast between the active and inactive gears, made the active gear "pop out" and visible in the peripheral view.

The visual feedback in VRA could be compared to how the gear selection can be visually perceived from the position of a polystable shifter. Perhaps moving this information into the instrument cluster could be one of the key elements in making monostable shifters safer. If the pattern is not shown for monostable shifters in a way that can easily be seen, people will have to create mental maps with a risk of making mistakes in stressful situations.

The attitudes toward the auditory feedback varied among the participants, some liked it while others didn't. Regardless of preference, there were general aspects that were common subjects of complaints. Just as suggested by Campbell et al. (2004), speech was perceived as annoying even by those who also thought it sounded kind. It is uncertain if the source of irritation was the speech itself or the frequency in which it was presented. The frequency was namely the second complaint, especially when people had to move through longer sequences (i.e. "P" to "D") and had to listen as the voice interrupted itself when it announced one abbreviation after another. Several participants mentioned that it reminded them of a computer that had technical problems and therefore made them worry that

the car might also have problems. In a realistic situation people would not change gears as often in ATs as they did in the usability test, but they might have to go several steps. To avoid the problems related to signal frequency, the VRA concept only included auditory signals for the gear positions that had been sources of confusion in the past, i.e. “P”, “R” and “D”. Since the number of auditory signals was less than five, earcons could be used instead of speech. An option of turning on and off the auditory feedback was included in VRA because of the individual differences and because it didn’t contribute to any significant changes in the usability.

6.2 RELEVANCE

Monostable shifters drastically alter the communication between the shifter and user. It sets demands on people as their habits are challenged and a new way of navigation must be learned for something that is an automated movement for drivers. With the changes in physical behavior it is important to analyze other components, such as feedback, to make the transition from polystable to monostable shifters safe and enjoyable. By not recognizing the communication differences between polystable and monostable shifters, designers risk developing systems that fail in situations where quick responses or automated actions are required.

6.3 REFLECTION

It was hard to determine if the usability of the monostable was improved to the point that its safety was comparable to polystable shifters. This would have been relevant information and a polystable shifter prototype should therefore have been included in the usability analysis. The monostable prototype shifters itself had an issue that might have affected the results. The sensors made a clicking sound whenever people made a gear selection. By doing so, all concepts received a type of auditory feedback and it is therefore hard to determine if the purely visual concept can be compared to the purely auditory one.

It was known, from the literature review, that earcons would have been a better choice than speech for gear selection feedback. Speech was chosen because of its comparability to the visual abbreviations thus allowing for a comparison between different modalities. However, it would have been interesting to include a usability study that looked closer into earcons and auditory icons. This would mean that learnability would need to be taken into consideration and that the auditory signals could only be compared to each other and not to visual signals as well.

There might have been too many shifting tasks in each sequence during the usability study. People shifted constantly and did not have time to forget the selected gear, making it hard to assess how the feedback would function in a realistic situation where fewer gear changes are made. The task assignment frequency could also have contributed to annoyance among participants.

Although the final VRA concept was almost identical to VA, it is hard to predict if the applied changes made VRA better or worse in terms of usability. A usability study that included VRA should have been made in order to assess the final concept.

6.4 CONCLUSIONS

The results from the thesis have shown that the usability of monostable shifters can be improved with feedback. However, it is uncertain if the proposed feedback concepts would lessen the accidents caused by monostable designs since there were few errors in the usability studies. The amount of off-road glances can be reduced by introducing sound to the shifter feedback and thus using another modality than the visual one. It must be taken into consideration that auditory signals can be perceived as distracting, which would diminish the positive effects of multimodality.

6.4.1 Project objective and aims

The aim of the thesis project was to study if the usability of monostable shifters could be improved through feedback, specifically visual, auditory and multimodal. The objective was to design a feedback concept that improved the effectiveness, efficiency and user satisfaction of a specific monostable shifter.

Concepts that provided permanent visual feedback improved the usability compared to the control concept that didn't include dynamic feedback. Based on this knowledge, the final concept VRA included permanent visual information and auditory feedback for "P", "R" and "D".

6.4.2 Research question 1: Visual, auditory or multimodal

The first research question was if visual, auditory or multimodal feedback was better than no feedback. The statistical analysis indicated that concepts V and VA were better than the control concept C. The good performance of V was supported by the Lundkvist et al. (2016), who showed that visual feedback performs better in vehicles, even though it uses the same modality as the driving task. On the other hand concept VA was supported by the Wickens Multi Resource model, according to which adding sound would relieve the visual modality. Concepts VT and A were seldom better than C. It appears that it was not the modality that contributed to the usability, but rather the duration and permanency of the feedback.

6.4.3 Research question 2: Best feedback in terms of usability

The second research question was what type of feedback was best in terms of effectiveness, efficiency and user satisfaction. There was no difference between the concepts, including the control concept C, in respect to effectiveness since few errors were made.

Concepts with permanent visual feedback (V and VA) were better than VT, A and C in terms of efficiency and user satisfaction. Always being able to see the chosen gear allowed people to quickly control their position, giving a sense of security and control. The complementary auditory feedback in VA confirmed the selection and made people more secure in their choice but could also be perceived as annoying. People started trusting the auditory information only by the end of the driving session. This could be contributed to habits since feedback is usually given as a visual stimuli in vehicles. Another reason might be that people perceived the visual information quicker than the auditory.

The results from the Acceptance scale, product reaction cards and the participant feedback indicated that the user satisfaction of VA depended both on the individuals and on the auditory signal itself. Individuals who preferred VA to V did

so because the complementary auditory signal helped them keep their gaze on the road. Some also described the sound as caring and they felt as if though it looked out for them, confirming that their selection was correct. Individuals who disliked VA explained that the sound was irritating, overwhelming and distracting

A closer look at the data from the usability studies indicated that VA tended to be slightly better than V when it came to performance and slightly worse in terms of user satisfaction. The main performance difference between the two concepts was the number of off-road glances, where VA performed better because people did not have to look away from the road to confirm their selection.

6.4.4 Research question 3: Effect of the permanent visual stimuli

The third research question looked at how the permanent representation of visual stimuli affected the performance compared to temporary feedback. The concepts with permanent visual information performed better than the temporary feedback in terms of efficiency and user satisfaction. Temporary feedback required more time to navigate and were perceived as more stressful, regardless of the modality. It appeared that when people couldn't identify the selected gear they had to visualize or remember the previous selection. In the cases of both A and VT they sometimes moved the shifter in random directions to find a reference point.

There was no significant difference in efficiency or user satisfaction between visual and auditory feedback if they were shown for a limited amount of time (VT and A). In many cases a temporary visual feedback (VT) was seen as more stressful than an auditory feedback that was presented for the same amount of time. The reason might be that visual stimuli is usually permanent, therefore when the reality does not meet the expectations from previous experiences people became stressed. Another reason could be that the iconic memory is shorter than the echoic which means that it could be forgotten more easily when other tasks that required attention were presented (i.e. driving). It was observed that people looked more confused and distracted when they made mistakes in auditory concepts than in visual concepts. It took longer time for them to correct their position. This might also be attributed to the echoic memory or to do the fact that they had to visualize the position of the gear since they could not see it. Regardless of why, this should be taken into consideration when designing auditory signals in cars to avoid accidents due to feedback distractions.

6.5 RECOMMENDATIONS

Monostable shifters do not have to be disregarded in vehicles. However the feedback design must be changed when the physical design is. By removing several stable point, the communication between the shifter and the user changes. Excluding the visual and haptic feedback from the physical shifter means that new redundant feedback must be added somewhere else. The drivers should not be expected to search for their gear selection or remember all the steps that need to be taken to make a correct selection.

It is recommended to include the shifter pattern in the instrument cluster for monostable shifters, since the physical shifter does not provide this information. Not only should it be included, but the chosen gear should "pop out" to minimize detection problems.

Although it was decided that the physical shifter was not to be altered, the results from the benchmarking and the information regarding the Gestalt laws implied that the "P" selection could be separate from the rest of the gears. The action in itself is different from the other gears since it symbolizes rest while the other gears symbolize motion. Since failure in choosing "P" and "R" were the main reasons for accidents, it is important to minimize the possibilities of an incorrect selection.

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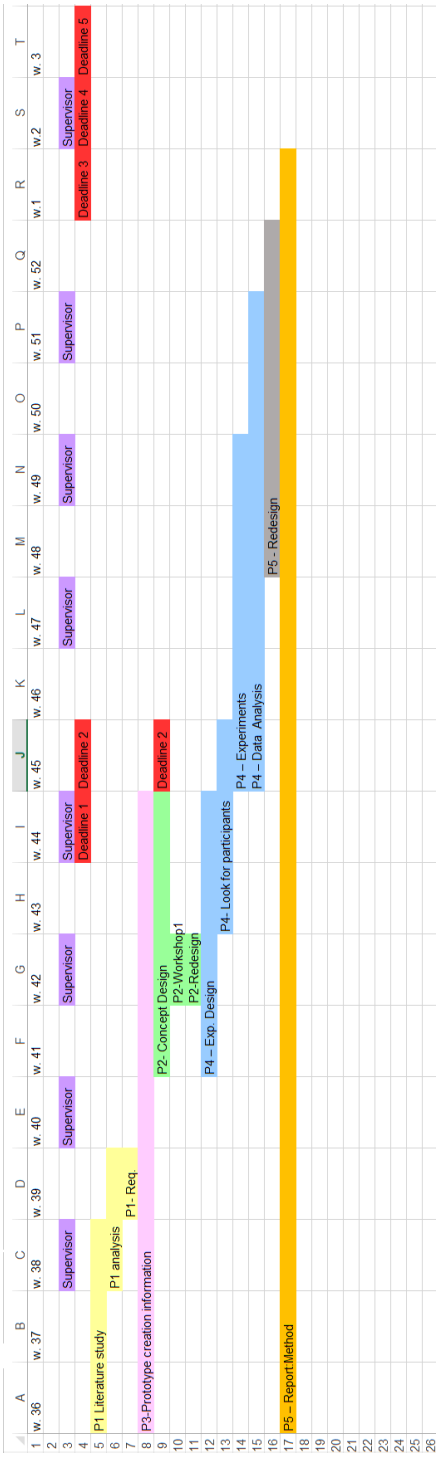
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APPENDIX A – GANTT CHART



APPENDIX B – MANUSCRIPT FOR THE USABILITY STUDY

Manus v.5

Förberedelser

A. Inför dagens pass

- Se över kameror (vMix Guide)
- Kontrollera att ljudet fungerar
- Pappersmuggar
- Godisskål med godis
- Penna
- Markeringspenna
- Se över att följande är utskrivet och finns:
  - Manus
  - Konceptbeskrivning
  - Tränings-lapp (1/TP)
  - Enkät-Användbarhet (5st/TP)
  - Samtyckesblankett (1/TP)
  - Enkät-Bakgrundsinformation (1/TP)
  - Enkät – Rangordning (1/TP)
  - TLX parvis jämförelse (1/TP)
  - NASA TLX förklaring
  - Product reaction (2/TP)

B. Inför varje TP (papper)

- Manus (placerad i mapp) - TL
- Konceptbeskrvining – TL
- 1 träningslapp
- 1 samryckesblankett
- 1 bakgrundsinformation-enkät
- 6st användbarhetsenkät
- 1 rangordnings-enkät
- 1 TLX parvis jämförelse
- 1 Product reaction (bäst)
- 1 Product reaction (sämst)
- 2st Product Reaction Cards

C. Inför varje TP (teknik)

- Kontrollera att det finns en förberedd mapp på E-Prime datorn för TP. Inuti den ska det finnas 1 mapp för varje koncept och 1 demomapp. Kontrollera att alla är i rätt ordning enligt Latin Square (01, 02, 03, 04, 05, 06)
- Kontrollera att väljaren fungerar och att den är i det automatiska läget (till höger)
- Kontrollera att mobiltelefonen med timern är laddad och att det finns hörlurar till TL.

D. Uppstart av labbet

- Starta och logga in på de tre datorerna
- Kontrollera att E-Prime datorn och instruktions-skärmen kommunicerar med varandra.
- Kontrollera att ljudet är på.
- Starta V-mix.
- Open pre-set (Tanyas examensarbete HT 16)
- Kontrollera att alla kameror fungerar och är fokuserade.
- OBS! Gå in i kugghjulet bredvid record och bekräfta inställningarna genom att trycka OK.
- I E-Prime datorn, klicka på mappen för testpersonen.
- Starta E-Prime demo run-filen som finns i TP:s mapp (01-demo). Se till att växelschemat och instruktionen visas.
- Ta fram timern med inställningen 13s 11 omgångar(Interval Tabata Timer), inga pauser emellan. Koppla i hörlurar i mobilen.

Manus för testet

Introduktion

- Hej och välkommen. Mitt namn är Tanya och jag kommer att vara din testledare idag. Testet som du kommer att delta i är en del av mitt examensarbete här på LTU.
- Du kan lägga dina saker här i bokhyllan eller på bordet.
- Kom ihåg att stänga av mobilen.
- Du kan sätta dig där vid ratten. Känn efter så att du sitter bekvämt.

START AV UPPLÄSNING HÄR:

- För att experimentet ska vara vetenskapligt behöver jag kontrollera att alla testpersoner får exakt samma information. Därför kommer jag att läsa innantill ur ett manus.
- Du kommer att få medverka i ett körtest som kommer att ta ca 1 timme. Det kommer att ingå 5 olika körbanor och efter varje bana kommer du att få fylla i en enkät. Jag kommer även att ställa några frågor.
- När du är färdig med att testa samtliga koncept kommer du att få fylla några fler enkäter.
- Om du av någon anledning skulle behöva avbryta så säger du till mig.

- Den här studien fokuserar på olika typer av feedback för en växelväljare i en automatväxlad bil. Med feedback menas hur valet av växel kommuniceras till dig som användare.
- De olika koncepten kommer att fungera på olika sätt. Därför är det viktigt att du är så ärlig som möjligt när du svarar på frågorna. Det är viktigt att du beskriver dina upplevelser, inte vad du tror att du borde ha upplevt.

- Ingenting av vad du säger eller gör här inne kommer att kunna kopplas till dig som person. Dina svar sparas anonymt. Jag är intresserad av att studera gruppen som helhet, inte enskilda individer.
- Det är också viktigt att du inte berättar för någon som ska vara testperson om vad du upplevde och tyckte och tänkte om experimentet, det kan nämligen påverka resultaten.

- Innan vi kör igång skulle jag vilja att du läser igenom och fyller i denna samtyckesblankett.
- SAMTYCKESBLANKETT
- Tack. Du kommer nu få svara på en enkät där jag ber dig att fylla i kort bakgrundsinformation om dig själv.
- ENKÄT - BAKGRUND

Genomgång av försöket

- Nu ska jag snabbt förklara upplägget i den här studien.
- Du kommer att använda en prototyp av en växelväljare, som du kan se till höger om dig.
- PEKA PÅ VÄLJAREN
- Denna kommer att användas för att utvärdera 5 olika feedback-koncept. Det är alltså inte den fysiska växelväljaren som utvärderas.
- Du kommer att använda prototypen samtidigt som du använder en bilsimulator.
- Vi kommer att börja med att bekanta oss med bilsimulatorn.

Förklaring av körsimulator

- I körsimuleringsprogrammet kommer du att köra på en trefilig väg där du ska vara uppmärksam på de vägskyltar som visas. Det kommer att se ut som på skärmen framför dig.
- Skyltarna visar när du ska byta fil och du ska lägga dig i den fil som pilen visar.
- Vilken fil tror du att du ska välja här?
- Det är viktigt att du är uppmärksam på skyltarna.
- Så fort du ser informationen på skyltarna ska du byta fil så snabbt men ändå behärskat som möjligt. Du ska alltså inte göra ett snyggt, glidande, filbyte.
- Totalt kommer du att köra 6 banor. Första banan kommer att vara en testkörning, så att du får bekanta dig med körningen och sedan startar testet.

- Du kommer att få börja med att köra igenom en kurva.
- Efter kurvan ska du lägga dig i mittenfilen.
- Jag kommer att hjälpa dig om du skulle hamna fel.
- Håll händerna på ratten under hela körtiden. Du behöver inte växla eller gasa.
- Är du redo att köra en övningsbana?
- TRYCK IGÅNG LCT-TEST
- TRYCK PÅ BANA 0
- TRYCK R EFTER SVÄNGEN
- AVSLUTA MED ESC

Förklaring av testupplägg

- Du kommer att köra en bana med varje feedback-koncept.
- Som du ser, på skärmen till vänster, kommer det att visas ett växelschema på den svarta delen och in instruktion på den gråa delen. Under tiden du kör kommer du att få instruktioner från denna skärm om vilken växel du ska välja.
- Det kommer att se ut och låta så här.
- TRYCK ENTER PÅ E-PRIME DATORN
- Du kan se den växeln som du har varit i och den som du ska välja.
- Efter en stund kommer nästa uppgift och då ser det ut så här.
- TRYCK ENTER PÅ E-PRIME DATORN
- Den växel som du nyss var i visas till vänster och till höger kan du se växeln du ska växla till.
- Vänta tills rösten har sagt klart vad du ska göra innan du väljer växel.
- Om du inte lyckas välja rätt växel innan nästa uppgift, så börjar du ändå nästa uppgift som om du hade valt rätt.
- Till exempel, om du inte hade lyckats välja M2 i föregående uppgift så påbörjar du ändå denna uppgift i M2.

Kamerastart

- Jag kommer nu att starta videoinspelningen. Videoinspelningen är endast till för en dataanalys. Jag kommer inte att använda inspelningen i något annat syfte.
- Klicka på record för att starta inspelningen. Recordknappen kommer att lysa röd.

Förklaring av växelväljaren

- Du kommer att köra samtidigt som du väljer körläge.
- Det finns 7 olika körläge: P, R, N, D, M1, M2 och M3.
- P betyder park(parkera), R betyder reverse(backa), N betyder neutral(frilägre), D betyder drive (kör). M står för manuellt läge och det finns tre manuella lägen: M1, M2 och M3.
- Växelväljaren är endast en prototyp och är inte kopplad till körsituationen i simulatorn.

- Jag kommer nu att förklara hur växelväljaren fungerar och du kommer att testa använda den.
- Vi börjar i läget D. För att gå från D till R drar du reglaget framåt 2 gånger. Prova göra det.
- Som du känner fjädrar reglaget sedan tillbaka till ett utgångsläge, R är dock fortfarande vald.

- När du är i R och sedan vill välja P, drar du reglaget framåt en gång.
- När du är i P och sedan vill välja D, drar du reglaget bakåt 3 gånger.
- Vill du gå från D till N drar du reglaget framåt en gång.
- Vill du gå från N till M2 drar du reglaget bakåt en gång, passerar D, och sedan till vänster. Som du märker stannar reglaget kvar i det manuella läget.
- För att gå från M2 till M1, drar du reglaget framåt en gång. Reglaget fjädrar tillbaka men M1 är fortfarande vald.
- För att gå från M1 till M3 drar du reglaget bakåt två gånger.
- Vill du gå från M3 till N, behöver du dra reglaget först till höger och sedan framåt 1 gång. Du behöver alltså inte gå genom M2 för att komma ut ur det manuella läget.
- Observera att om du försöker välja M2 från ett annat läge än D får du en varning. Testa göra det, dra spaken till vänster. Dra spaken tillbaka till höger för att återgå till det läget du var innan du fick varningen, alltså N.

Träning

- Du kommer nu att få öva på att använda växelväljaren innan vi fortsätter med själva testet.
- Inga feedback-koncept testas nu. Skulle du välja fel så säger jag till.

Du börjar i läget P.

- TRÄNINGSBLANKETT
- Nu när du har bekantat dig med växelväljaren kan vi börja med testerna.
- Avsluta E-Prime genom: ALT+CTRL+SHIFT

Omgång 1 (A)

- Kontrollera att spaken är i A-läget OBS, VIKTIGT!!
- Klicka på Mappen för det första konceptet (02) och starta E-Prime Run filen. Stanna kvar i fönstret där namnet på konceptet visas.
- Du ska nu testa det första konceptet.
- Tryck Enter så att växelschemat visas.
- LÄS UR KONCEPTBESKRIVNING.

- Under testet vill jag att du är så koncentrerad på testet som möjligt. Har du några frågor så går det bra att ställa dem nu.
- Håll händerna på ratten under hela körtiden förutom då du växlar. Kom ihåg att växla först när instruktionsrösten är färdig.
- Är du redo att starta?
- LCT TEST, BANA 1
- TRYCK R OM TP HAMNAR FEL.
- Starta Timern precis innan Start-skylten. Den räknar ner till 3.
- Tryck på Enter i E-Prime datorn varje gång timer säger till.
- ESC FÖR ATT GÅ UR

- Vad tyckte du om det här konceptet? Hade du hjälp av feedbacken?
  - Om de svarar ja: Hur hjälpte den dig
  - Om KVA: Hur använde du den visuella informationen? Hur använde du den auditiva informationen?
- Du kommer nu att få fylla i en enkät. Fundera inte så länge på dina svar, utan svara med din spontana känsla.
- Användarvänlighet-enkät för konceptet
- Observera att frågorna handlar om feedbacken, inte den fysiska väljaren.

Om konceptet ska ha Product Reaction Cards

- Du kommer nu att få ett paper som innehåller en mängd olika ord.
- Välj de orden som stämmer bäst överens med din upplevelse av konceptet.
- Ge Product Reaction –enkät

- Efter att de är klara. Ge de en markeringspenna.
- Välj nu ut 5 av de valda orden som du anser stämmer mest överens med din upplevelse av konceptet.

Omgång 2 (A)

- Kontrollera att spaken är i A-läget OBS, VIKTIGT!!
- Klicka på Mappen för det första konceptet (03) och starta E-Prime Run filen. Stanna kvar i fönstret där namnet på konceptet visas.
- Du ska nu testa det andra konceptet.
- Tryck Enter så att växelschemat visas.
- LÄS UR KONCEPTBESKRIVNING

- Under testet vill jag att du är så koncentrerad på testet som möjligt. Har du några frågor så går det bra att ställa dem nu.
- Håll händerna på ratten under hela körtiden förutom då du växlar. Kom ihåg att växla först när instruktionsrösten är färdig.
- Är du redo att starta?
- LCT TEST, BANA 2
- TRYCK R OM TP HAMNAR FEL.
- Starta Timern precis innan Start-skylten. Den räknar ner till 3.
- Tryck på Enter i E-Prime datorn varje gång timer säger till.
- ESC FÖR ATT GÅ UR

- Vad tyckte du om det här konceptet? Hade du hjälp av feedbacken?
  - Om de svarar ja: Hur hjälpte den dig
  - Om KVA: Hur använde du den visuella informationen? Hur använde du den auditiva informationen?
- Du kommer nu att få fylla i en enkät. Fundera inte så länge på dina svar, utan svara med din spontana känsla.
- Användarvänlighet-enkät för konceptet
- Observera att frågorna handlar om feedbacken, inte den fysiska väljaren.

Om konceptet ska ha Product Reaction Cards

- Du kommer nu att få ett paper som innehåller en mängd olika ord.
- Välj de orden som stämmer bäst överens med din upplevelse av konceptet.
- Ge Product Reaction –enkät

- Efter att de är klara. Ge de en markeringspenna.
- Välj nu ut 5 av de valda orden som du anser stämmer mest överens med din upplevelse av konceptet.

Omgång 3 (A)

- Kontrollera att spaken är i A-läget OBS, VIKTIGT!!
- Klicka på Mappen för det första konceptet (04) och starta E-Prime Run filen. Stanna kvar i fönstret där namnet på konceptet visas.
- Du ska nu testa det tredje konceptet.
- Tryck Enter så att växelschemat visas.
- LÄS UR KONCEPTBESKRIVNING

- Under testet vill jag att du är så koncentrerad på testet som möjligt. Har du några frågor så går det bra att ställa dem nu.
- Håll händerna på ratten under hela körtiden förutom då du växlar. Kom ihåg att växla först när instruktionsrösten är färdig.
- Är du redo att starta?
- LCT TEST, BANA 3
- TRYCK R OM TP HAMNAR FEL.
- Starta Timern precis innan Start-skylten. Den räknar ner till 3.
- Tryck på Enter i E-Prime datorn varje gång timer säger till.
- TRYCK ESC FÖR ATT GÅ UR

- Vad tyckte du om det här konceptet? Hade du hjälp av feedbacken?
  - Om de svarar ja: Hur hjälpte den dig
  - Om KVA: Hur använde du den visuella informationen? Hur använde du den auditiva informationen?
- Du kommer nu att få fylla i en enkät. Fundera inte så länge på dina svar, utan svara med din spontana känsla.
- Användarvänlighet-enkät för konceptet
- Observera att frågorna handlar om feedbacken, inte den fysiska väljaren.

Om konceptet ska ha Product Reaction Cards

- Du kommer nu att få ett paper som innehåller en mängd olika ord.
- Välj de orden som stämmer bäst överens med din upplevelse av konceptet.
- Ge Product Reaction –enkät

- Efter att de är klara. Ge de en markeringspenna.
- Välj nu ut 5 av de valda orden som du anser stämmer mest överens med din upplevelse av konceptet.

Omgång 4 (M)

- Kontrollera att spaken är i M-läget OBS, VIKTIGT!!
- Klicka på Mappen för det första konceptet (05) och starta E-Prime Run filen. Stanna kvar i fönstret där namnet på konceptet visas
- Du ska nu testa det fjärde konceptet.
- Tryck Enter så att växelschemat visas.
- LÄS UR KONCEPTBESKRIVNING

- Under testet vill jag att du är så koncentrerad på testet som möjligt. Har du några frågor så går det bra att ställa dem nu.
- Håll händerna på ratten under hela körtiden förutom då du växlar. Kom ihåg att växla först när instruktionsrösten är färdig.
- Är du redo att starta?
- LCT TEST, BANA 4
- TRYCK R OM TP HAMNAR FEL.
- Starta Timern precis innan Start-skylten. Den räknar ner till 3.
- Tryck på Enter i E-Prime datorn varje gång timer säger till.
- TRYCK ESC FÖR ATT GÅ UR

- Vad tyckte du om det här konceptet? Hade du hjälp av feedbacken?
  - Om de svarar ja: Hur hjälpte den dig
  - Om KVA: Hur använde du den visuella informationen? Hur använde du den auditiva informationen?
- Du kommer nu att få fylla i en enkät. Fundera inte så länge på dina svar, utan svara med din spontana känsla.
- Användarvänlighet-enkät för konceptet
- Observera att frågorna handlar om feedbacken, inte den fysiska väljaren.

Om konceptet ska ha Product Reaction Cards

- Du kommer nu att få ett paper som innehåller en mängd olika ord.
- Välj de orden som stämmer bäst överens med din upplevelse av konceptet.
- Ge Product Reaction –enkät

- Efter att de är klara. Ge de en markeringspenna.
- Välj nu ut 5 av de valda orden som du anser stämmer mest överens med din upplevelse av konceptet.

Omgång 5 (A)

- Kontrollera att spaken är i A-läget OBS, VIKTIGT!!
- Klicka på Mappen för det första konceptet (06) och starta E-Prime Run filen. Stanna kvar i fönstret där namnet på konceptet visas
- Du ska nu testa det femte och sista konceptet.
- Tryck Enter så att växelschemat visas.
- LÄS UR KONCEPTBESKRIVNING

- Under testet vill jag att du är så koncentrerad på testet som möjligt. Har du några frågor så går det bra att ställa dem nu.
- Håll händerna på ratten under hela körtiden förutom då du växlar. Kom ihåg att växla först när instruktionsrösten är färdig.
- Är du redo att starta?
- LCT TEST, BANA 5
- TRYCK R OM TP HAMNAR FEL.
- Starta Timern precis innan Start-skylten. Den räknar ner till 3.
- Tryck på Enter i E-Prime datorn varje gång timer säger till.
- TRYCK ESC FÖR ATT GÅ UR

- Vad tyckte du om det här konceptet? Hade du hjälp av feedbacken?
  - Om de svarar ja: Hur hjälpte den dig
  - Om KVA: Hur använde du den visuella informationen? Hur använde du den auditiva informationen?
- Du kommer nu att få fylla i en enkät. Fundera inte så länge på dina svar, utan svara med din spontana känsla.
- Användarvänlighet-enkät för konceptet
- Observera att frågorna handlar om feedbacken, inte den fysiska väljaren.

Om konceptet ska ha Product Reaction Cards

- Du kommer nu att få ett paper som innehåller en mängd olika ord.
- Välj de orden som stämmer bäst överens med din upplevelse av konceptet.
- Ge Product Reaction –enkät

- Efter att de är klara. Ge de en markeringspenna.
- Välj nu ut 5 av de valda orden som du anser stämmer mest överens med din upplevelse av konceptet.

Utvärdering

- Efter varje koncept fick du fylla i en skattningsskala som bestod av 6 olika faktorer för att uppskatta din arbetsbelastning.
- Du kommer nu att få ett papper som innehåller 15 rutor.
- Visa pappret
- Varje ruta innehåller två ord, dessa är namnen på skalorna som du fyllde i.
- Peka på en ruta och visa att det är 2 ord
- Jag vill att du, i varje ruta, väljer den faktorn som var viktigast för din upplevelse av arbetsbelastningen i uppgifterna.
- Med uppgift menas att välja växel. Det är alltså inte kopplat till något specifikt koncept utan handlar om självaste uppgiften att välja växel.
- Det finns inget korrekt mönster eller svar. Jag är intresserad av dina åsikter.
- Ge JÄMFÖRELSELAPPAR
- Du får även en fusklapp som förklarar vad varje faktor betyder.
- Ge fusklappen

- Tack. Till sist skulle jag vilja att du rangordnar koncepten med en siffra från 1 till 5, där 5 är bäst och 1 är sämst. Varje siffra får endast användas 1 gång.
- Du får gärna berätta hur du tänker.
- RANGORDNING-Enkät

Avslutande

- Tack. Jag ska nu avsluta inspelningen.
- STÄNG AV VIDEO

- Stort tack för ditt deltagande! Ditt bidrag är till stor nytta och jag skulle gärna vilja ge dig ett presentkort som ett litet tack.
- KUPONG

- Kom ihåg att inte berätta om testet och vad du tyckte om reglaget för någon så länge testet är igång, vilket är till början av januari 2017. Det kan nämligen påverka resultaten.

Efter att de har gått

- I testpersonens mapp i E-Prime datorn skapa ett enkelt .txt dokument som du döper till TPXX-Sammanfattning. XX står för TP:s nummer.
- I dokumentet antecknar du snabbt vad TP hade att säga om de olika koncepten och ifall du observerade något speciellt.

- I LCT datorn: Lägg TP:s LCT filer i TP:s mapp. Ta bort den första filen som är för träningen.
- I Vmix datorn: Lägg TP:s filmer i TP:s mapp.

- Samla ihop alla papper som tillhör TP och lägg dem i en plastficka. Sätt in plastfickan i en pärm.

- Se till att väljaren är i A.
- Förbered papperna för nästa TP.

APPENDIX C – TRAINING SHEET FOR THE SHIFTER

		Y	N
P -> R	-1		
R -> D	-2		
D - > M2	U		
M2 -> M1	+1		
M1 -> M3	-2		
M3 -> N	N, +1		
N -> R	+1		
R -> M2	-2, U		
M2 -> M3	-1		
M3 -> P	N, +3		

		Y	N
P -> R	-1		
R -> D	-2		
D - > M2	U		
M2 -> M1	+1		
M1 -> M3	-2		
M3 -> N	N, +1		
N -> R	+1		
R -> M2	-2, U		
M2 -> M3	-1		
M3 -> P	N, +3		

		Y	N
P -> R	-1		
R -> D	-2		
D - > M2	U		
M2 -> M1	+1		
M1 -> M3	-2		
M3 -> N	N, +1		
N -> R	+1		
R -> M2	-2, U		
M2 -> M3	-1		
M3 -> P	N, +3		

P

R

M1 N

M2---------- D

M3

APPENDIX D – TASK SEQUENCES FOR THE USABILITY STUDY

The usability study included a total of 27 different tasks. The complexity level of each task was determined by the amount of steps that the shifter required to achieve a required gear from the previously selected one. The matrix to the left of the task list shows how many steps are required to choose the gear shown in the horizontal axis if the gears in the vertical are chosen. This information was used to design the task sequences so that each sequence included approximately the same amount of steps (22+/-1) and complexity level.

	P	R	N	D	M(2)	M3	M1
P		1	2	3	4	5	5
R	1		1	2	3	4	4
N	2	1		1	2	3	3
D	3	2	1		1	2	2
M(2)	4	3	2	1		1	1
M3	4	3	2	1	1		2
M1	4	3	2	1	1	2	

Task	steps	Command
Task 1	1	PR
Task 2	2	PN
Task 3	3	PD
Task 4	4	PM2
Task 5	1	RP
Task 6	2	RD
Task 7	3	RM2
Task 8	1	NR
Task 9	2	NP
Task 10	2	NM2
Task 11	3	NM3
Task 12	3	NM1
Task 13	3	DP
Task 14	2	DR
Task 15	1	DN
Task 16	1	DM2
Task 17	2	DM1
Task 18	3	M2R
Task 19	2	M2N
Task 20	1	M2D
Task 21	1	M2M1
Task 22	4	M1P
Task 23	3	M1R
Task 24	1	M1D
Task 25	2	M3N
Task 26	1	M3M2
Task 27	2	M1M3

Task order	S 1	S2	S3	S4	S5
1	Task 1	Task 5	Task 15	Task 21	Task 8
2	Task 7	Task 2	Task 11	Task 23	Task 6
3	Task 19	Task 11	Task 25	Task 6	Task 13
4	Task 9	Task 26	Task 9	Task 13	Task 4
5	Task 3	Task 20	Task 4	Task 4	Task 19
6	Task 17	Task 13	Task 20	Task 19	Task 12
7	Task 22	Task 4	Task 14	Task 8	Task 24
8	Task 2	Task 18	Task 5	Task 5	Task 14
9	Task 8	Task 6	Task 3	Task 2	Task 5
10	Task 6	Task 15	Task 16	Task 10	Task 3
11	Task 16	Task 8	Task 18	Task 20	Task 16
Sum (steps)	23	22	23	22	23

APPENDIX E – LATIN SQUARE FOR THE CONCEPT ORDER

Latin Square used to decide the order in which to present the concepts to the participants. By following the Latin Square the concepts were counter balanced and each concepts was tested with all the possible task sequences. The yellow blocks indicate that the marked concept was evaluated with a product reaction card by the TP.

	S1	S2	S3	S4	S5
TP 1	C1 = C	C2 = V	C5 = VA	C3 = A	C4 = VT
TP 2	C2 = V	C3 = A	C1 = C	C4 = VT	C5 = VA
TP 3	C3 = A	C4 = VT	C2 = V	C5 = VA	C1 = C
TP 4	C4 = VT	C5 = VA	C3 = A	C1 = C	C2 = V
TP 5	C5 = VA	C1 = C	C4 = VT	C2 = V	C3 = A

	S1	S2	S3	S4	S5
TP 6	C1 = C	C2 = V	C5 = VA	C3 = A	C4 = VT
TP 7	C2 = V	C3 = A	C1 = C	C4 = VT	C5 = VA
TP 8	C3 = A	C4 = VT	C2 = V	C5 = VA	C1 = C
TP 9	C4 = VT	C5 = VA	C3 = A	C1 = C	C2 = V
TP 10	C5 = VA	C1 = C	C4 = VT	C2 = V	C3 = A

	S1	S2	S3	S4	S5
TP 11	C1 = C	C2 = V	C5 = VA	C3 = A	C4 = VT
TP 12	C2 = V	C3 = A	C1 = C	C4 = VT	C5 = VA
TP 13	C3 = A	C4 = VT	C2 = V	C5 = VA	C1 = C
TP 14	C4 = VT	C5 = VA	C3 = A	C1 = C	C2 = V
TP 15	C5 = VA	C1 = C	C4 = VT	C2 = V	C3 = A

	S1	S2	S3	S4	S5
TP 16	C1 = C	C2 = V	C5 = VA	C3 = A	C4 = VT
TP 17	C2 = V	C3 = A	C1 = C	C4 = VT	C5 = VA
TP 18	C3 = A	C4 = VT	C2 = V	C5 = VA	C1 = C
TP 19	C4 = VT	C5 = VA	C3 = A	C1 = C	C2 = V
TP 20	C5 = VA	C1 = C	C4 = VT	C2 = V	C3 = A

	S1	S2	S3	S4	S5
TP 21	C1 = C	C2 = V	C5 = VA	C3 = A	C4 = VT
TP 22	C2 = V	C3 = A	C1 = C	C4 = VT	C5 = VA
TP 23	C3 = A	C4 = VT	C2 = V	C5 = VA	C1 = C
TP 24	C4 = VT	C5 = VA	C3 = A	C1 = C	C2 = V
TP 25	C5 = VA	C1 = C	C4 = VT	C2 = V	C3 = A

APPENDIX F – CONSENT FORM

SAMTYCKE OM DELTAGANDE I STUDIE



Studien är en del av ett examensarbete inom Teknisk design på Luleå Tekniska Universitet.

Syftet med studien är att undersöka användbarheten hos olika feedback-koncept för en växelväljare.

Moment: Studien tar ca 1 timme och innehåller följande moment:

- Välja olika körlägen enligt instruktioner.
- Köra ett standardiserat körsimuleringsprogram, Lane Change Test.
- Fylla i enkäter
- Svara på frågor
- Rangordna koncepten utifrån användbarhet

Frivilligt deltagande: Deltagande i studien sker på frivillig basis och som testperson har du rätt att när som helst avbryta din medverkan.

Insamlat datamaterial: Videoinspelat material, enkätsvar och intervjusvar kommer enbart att användas vid analys av data. Materialet kommer inte att spridas eller visas utanför studien.

Jag samtycker om deltagande i studien:

Luleå Datum: _____

Namnteckning: _____

Namnförtydligande: _____

APPENDIX G – BACKGROUND INFORMATION

Testperson: _____

Bakgrundsinformation

- Kvinna
- Man

- Ålder: _____

1. Hur länge (år) har du haft körkort? _____

2. Hur ofta kör du bil?

- Aldrig
- Någon gång per år
- Någon gång per månad
- Någon gång per vecka
- Nästan dagligen
- Flera gånger per dagen

3. Hur ofta kör du med en automatväxlad bil (R-N-D)?

- Aldrig
- Någon gång per år
- Någon gång per månad
- Någon gång per vecka
- Nästan dagligen
- Flera gånger per dag

4. Hur ofta spelar du dator/tv-spel (ej mobilspel)?

- Aldrig
- Någon gång per år
- Någon gång per månad
- Någon gång per vecka
- Nästan dagligen
- Flera gånger per dag



5. Har du använt en körsimulator förut?

- Ja
- Nej
- Vet ej

6. Hur intresserad är du av att använda ny teknik?

Inte alls Mycket

7. Hur frustrerad kan du bli när du försöker lära dig ny teknik?

Inte alls Mycket

APPENDIX H – POST TASK SURVEYS, ACCEPTANCE SCALE AND NASA TLX

Testperson: _____

Enkät: Användarvänlighet, Koncept C



Har du använt en växelväljare med en liknande typ av feedback?

- ○ Nej
- ○ Ja, sällan
- ○ Ja, ofta

Din upplevelse av feedback-konceptet är…

1	Användbar						Meningslös
2	Trevlig						Otrevlig
3	Dålig						Bra
4	Rolig						Tråkig
5	Effektiv						Ineffektiv
6	Irriterande						Angenäm
7	Behjälplig						Usel
8	Oönskad						Önskad
9	Uppmärksamhetshöjande						Sövande



Arbetsbelastning, skattningsskalor

Markera på skalen det som motsvarar din upplevelse.

1. Mentala krav

Hur mycket behövde du tänka, söka, beräkna, komma ihåg etc.?



Väldigt lite Väldigt mycket

2. Fysiska krav

Hur mycket behövde du trycka, dra, vrida, kontrollera, aktivera etc.?



Väldigt lite Väldigt mycket

3. Tidskrav

Hur mycket tidspress kände du på grund av hastighet som uppgiften skedde?



Väldigt lite Väldigt mycket

4. Prestation

Hur framgångsrik tycker du att du var i att åstadkomma uppgiftens mål?



Perfekt Misslyckad

5. Ansträngning

Hur hårt var du tvungen att arbeta för att åstadkomma din prestation?



Väldigt lite Väldigt mycket

6. Frustration

Hur osäker, nedslagen, stressad och irriterad kände du dig under uppgiften?



Väldigt lite Väldigt mycket

Testperson: _____

Enkät: Användarvänlighet, Koncept V



Har du använt en växelväljare med en liknande typ av feedback?

- Nej
- Ja, sällan
- Ja, ofta

Din upplevelse av feedback-konceptet är…

1	Användbar						Meningslös
2	Trevlig						Otrevlig
3	Dålig						Bra
4	Rolig						Tråkig
5	Effektiv						Ineffektiv
6	Irriterande						Angenäm
7	Behjälplig						Usel
8	Oönskad						Önskad
9	Uppmärksamhetshöjande						Sövande



Arbetsbelastning, skattningsskalor

Markera på skalen det som motsvarar din upplevelse.

1. Mentala krav

Hur mycket behövde du tänka, söka, beräkna, komma ihåg etc.?



Väldigt lite Väldigt mycket

2. Fysiska krav

Hur mycket behövde du trycka, dra, vrida, kontrollera, aktivera etc.?



Väldigt lite Väldigt mycket

3. Tidskrav

Hur mycket tidspress kände du på grund av hastighet som uppgiften skedde?



Väldigt lite Väldigt mycket

4. Prestation

Hur framgångsrik tycker du att du var i att åstadkomma uppgiftens mål?



Perfekt Misslyckad

5. Ansträngning

Hur hårt var du tvungen att arbeta för att åstadkomma din prestation?



Väldigt lite Väldigt mycket

6. Frustration

Hur osäker, nedslagen, stressad och irriterad kände du dig under uppgiften?



Väldigt lite Väldigt mycket

Testperson: _____

Enkät: Användarvänlighet, Koncept A



Har du använt en växelväljare med en liknande typ av feedback?

- Nej
- Ja, sällan
- Ja, ofta

Din upplevelse av feedback-konceptet är…

1	Användbar						Meningslös
2	Trevlig						Otrevlig
3	Dålig						Bra
4	Rolig						Tråkig
5	Effektiv						Ineffektiv
6	Irriterande						Angenäm
7	Behjälplig						Usel
8	Oönskad						Önskad
9	Uppmärksamhetshöjande						Sövande



Arbetsbelastning, skattningsskalor

Markera på skalen det som motsvarar din upplevelse.

1. Mentala krav

Hur mycket behövde du tänka, söka, beräkna, komma ihåg etc.?



Väldigt lite Väldigt mycket

2. Fysiska krav

Hur mycket behövde du trycka, dra, vrida, kontrollera, aktivera etc.?



Väldigt lite Väldigt mycket

3. Tidskrav

Hur mycket tidspress kände du på grund av hastighet som uppgiften skedde?



Väldigt lite Väldigt mycket

4. Prestation

Hur framgångsrik tycker du att du var i att åstadkomma uppgiftens mål?



Perfekt Misslyckad

5. Ansträngning

Hur hårt var du tvungen att arbeta för att åstadkomma din prestation?



Väldigt lite Väldigt mycket

6. Frustration

Hur osäker, nedslagen, stressad och irriterad kände du dig under uppgiften?



Väldigt lite Väldigt mycket

Testperson: _____

Enkät: Användarvänlighet, Koncept VT



Har du använt en växelväljare med en liknande typ av feedback?

- Nej
- Ja, sällan
- Ja, ofta

Din upplevelse av feedback-konceptet är…

1	Användbar						Meningslös
2	Trevlig						Otrevlig
3	Dålig						Bra
4	Rolig						Tråkig
5	Effektiv						Ineffektiv
6	Irriterande						Angenäm
7	Behjälplig						Usel
8	Oönskad						Önskad
9	Uppmärksamhetshöjande						Sövande



Arbetsbelastning, skattningsskalor

Markera på skalen det som motsvarar din upplevelse.

1. Mentala krav

Hur mycket behövde du tänka, söka, beräkna, komma ihåg etc.?



Väldigt lite Väldigt mycket

2. Fysiska krav

Hur mycket behövde du trycka, dra, vrida, kontrollera, aktivera etc.?



Väldigt lite Väldigt mycket

3. Tidskrav

Hur mycket tidspress kände du på grund av hastighet som uppgiften skedde?



Väldigt lite Väldigt mycket

4. Prestation

Hur framgångsrik tycker du att du var i att åstadkomma uppgiftens mål?



Perfekt Misslyckad

5. Ansträngning

Hur hårt var du tvungen att arbeta för att åstadkomma din prestation?



Väldigt lite Väldigt mycket

6. Frustration

Hur osäker, nedslagen, stressad och irriterad kände du dig under uppgiften?



Väldigt lite Väldigt mycket

Testperson: _____

Enkät: Användarvänlighet, Koncept VA



Har du använt en växelväljare med en liknande typ av feedback?

- Nej
- Ja, sällan
- Ja, ofta

Din upplevelse av feedback-konceptet är…

1	Användbar						Meningslös
2	Trevlig						Otrevlig
3	Dålig						Bra
4	Rolig						Tråkig
5	Effektiv						Ineffektiv
6	Irriterande						Angenäm
7	Behjälplig						Usel
8	Oönskad						Önskad
9	Uppmärksamhetshöjande						Sövande



Arbetsbelastning, skattningsskalor

Markera på skalen det som motsvarar din upplevelse.

1. Mentala krav

Hur mycket behövde du tänka, söka, beräkna, komma ihåg etc.?



Väldigt lite Väldigt mycket

2. Fysiska krav

Hur mycket behövde du trycka, dra, vrida, kontrollera, aktivera etc.?



Väldigt lite Väldigt mycket

3. Tidskrav

Hur mycket tidspress kände du på grund av hastighet som uppgiften skedde?



Väldigt lite Väldigt mycket

4. Prestation

Hur framgångsrik tycker du att du var i att åstadkomma uppgiftens mål?



Perfekt Misslyckad

5. Ansträngning

Hur hårt var du tvungen att arbeta för att åstadkomma din prestation?



Väldigt lite Väldigt mycket

6. Frustration

Hur osäker, nedslagen, stressad och irriterad kände du dig under uppgiften?



Väldigt lite Väldigt mycket

APPENDIX I – NASA TLX PAIRWISE COMPARISON

Testperson: _____

Parvis jämförelse

Välj den faktorn som var viktigast för din upplevelse av arbetsbelastningen. Gör detta för alla rutor.

Ansträngning eller Prestation	Tidskrav eller Frustration	Tidskrav eller Ansträngning
Fysiska krav eller Frustration	Prestation eller Frustration	Fysiska krav eller Tidskrav
Fysiska krav eller Prestation	Tidskrav eller Mentala krav	Frustration eller Ansträngning
Prestation eller Mentala krav	Prestation eller Tidskrav	Mentala krav eller Ansträngning
Mentala krav eller Fysiska krav	Ansträngning eller Fysiska krav	Frustration eller Mentala krav

APPENDIX J – PRODUCT REACTION CARDS

Testperson: _____ Koncept_____

Upplevelse av Konceptet

Välj de orden som stämmer bäst överens med din upplevelse av konceptet.

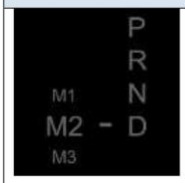
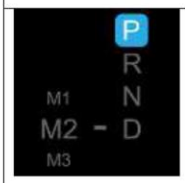
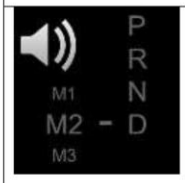
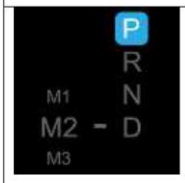
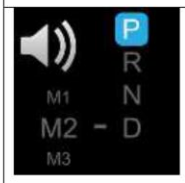
Allvarlig	Irriterande	Pålitlig
Arrogant	Justerbar	Relevant
Attraktiv	Kompatibel	Rolig
Begriplig	Komplicerad	Rättfram
Bekant	Konsekvent	Snabb
Bekväm	Kontrollerbar	Sofistikerad
Distraherande	Kreativ	Stabil
Dålig kvalitet	Lugn	Steril
Effektiv	Långsam	Stilren
Energisk	Meningsfull	Stressig
Engagerande	Motiverande	Stökig
Enkel	Mäktig	Störande
Enkel att använda	Nedlåtande	Svår
Entusiastisk	Ny	Tidskrävande
Exceptionell	Nytänkande	Tidssparande
Flexibel	Oattraktiv	Tillgänglig
Frustrerande	Obegriplig	Tilltalande
Fräsch	Oförutsägbar	Trygg
För teknisk	Ohjälpsam	Tråkig
Förutsägbar	Okontrollerbar	Underhållande
Förvirrande	Omfattande	Uppiggande
Förväntad	Opersonlig	Utdaterad
Gammal	Optimistisk	Vänlig
Hjälpsam	Ordinär	Värdelös
Bra kvalitet	Organiserad	Väsentlig
I vägen	Originell	Ömtålig
Imponerande	Osäker	Önskvärd
Inkonsekvent	Otillgänglig	Övertygande
Innovativ	Oönskad	Överväldigande
Inspirerande	Personlig	

APPENDIX K – RANKING SHEET

Testperson: _____

Rankning

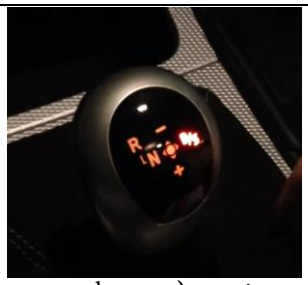
Betygsätt varje koncept med en siffra 1-5, där **5 är bäst och 1 är sämst**.
Varje siffra får endast användas 1 gång.

Koncept		Betyg (1—5)
	Koncept C Ingen feedback	
	Koncept V Visuell feedback (stannar)	
	Koncept A Auditiv feedback	
 	Koncept VT Temporär visuell feedback	
	Koncept VA Visuell och auditiv feedback	

APPENDIX L – BENCHMARKING

Shifter/car	Visual feedback	Auditory feedback	Tactile feedback
ZF E-Shifter Fiat Chrysler: Dodge Charger, Chrysler 300, Jeep Grand Cherokee (Monostable)	Abbreviations on the knob. Light intensity contrast. **Shifting pattern on instrument cluster, little contrast**.	Before update: chime when the door opens while not in P.	Feel how far the shifter moves, springs back after selection.
Steptronic BMW 5-series (Monostable R, N, D)	Abbreviations on the knob. Light intensity and color contrast. Green dot next to the chosen gear. P is not shown on the shift pattern unless it is chosen. Instrument cluster.	None	Feel how far the shifter moves, springs back after selection. Can feel if in cruiser or automatic mode. Feel when P is pressed.
DCT BMW M5 (Monostable)	Abbreviations on the knob. Light intensity contrast. Physical shifter stays in chosen position. Instrument cluster.	None	Feel how far the shifter moves and its position. Feel when passing through gears because of the small stops on the way. Feel how the buttons are pressed.
2016 Acura TLX (Button Shifter)	Abbreviations on the buttons form the shift pattern. Each gear has a differently shaped button. Active choice has a colored outline. P has a red active color, all other have green. Instrument cluster.	None	Change in texture, shape, position and angle. R is pulled backwards and D is placed on an angle that requires the driver to press forward.
Lincoln MKZ (Button Shifter)	Abbreviations on the buttons form the shift pattern. Color contrast, red is the active color. **Shifting pattern on the instrument cluster**.	None	Feel how the buttons are pressed.
Nissan Leaf (Monostable)	Abbreviations next to the knob. Red dot next to the active selection. Instrument cluster.	Beep in R.	Feel the direction in which the shifter moves. Feel when the P button is pressed.
Toyota Prius (Monostable)	Abbreviations on the knob. A green dot is shown only when P is chosen. Instrument cluster.	Beep in R.	Feel the direction in which the shifter moves. Feel when the P button is pressed.

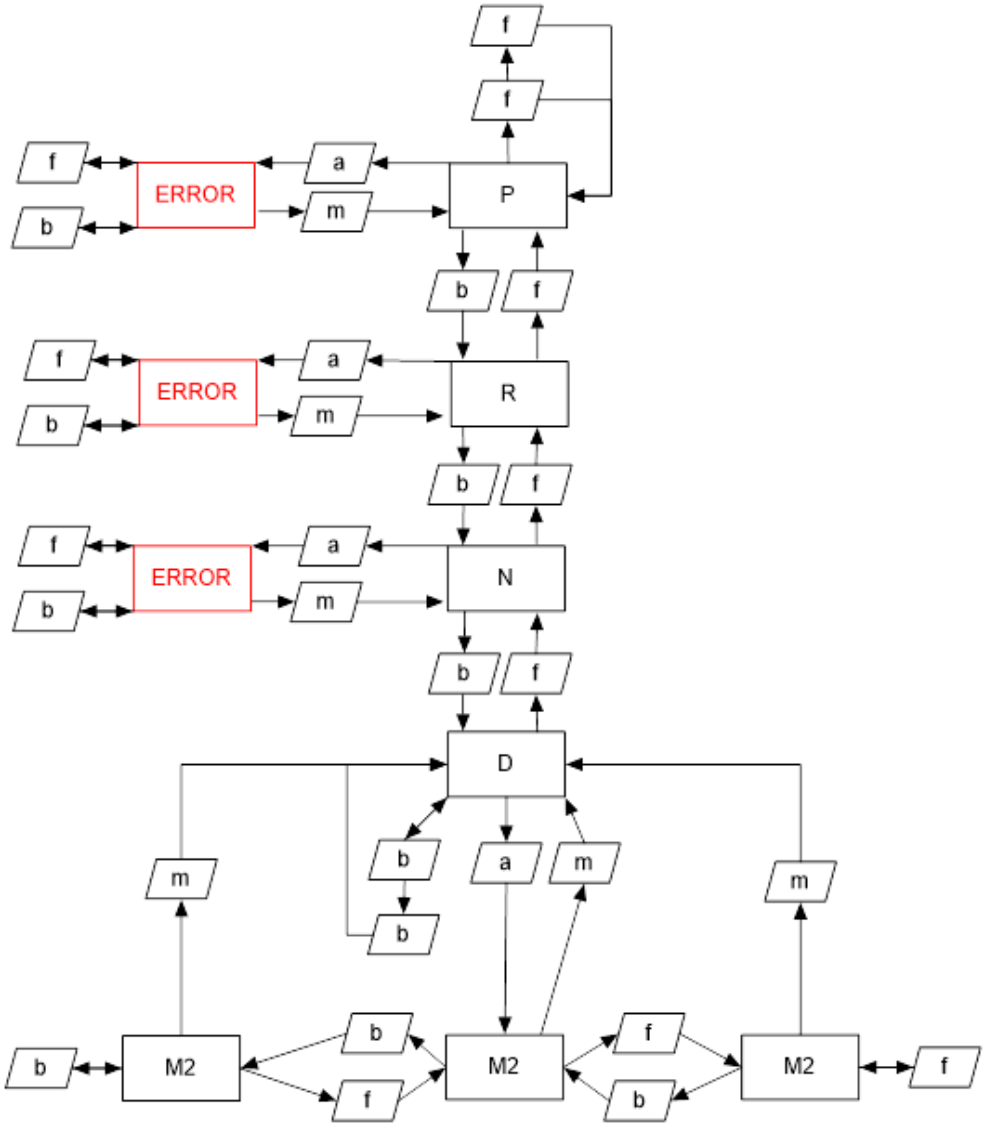
Shifter/car	Visual feedback	Auditory feedback	Tactile feedback
BMW i3 (Paddle on the wheel)	Shifting pattern is always shown (not in P). Green color for the chosen gear.	None.	Feel the movement.
7G-DCT. Mercedes Benz. (Paddle on the wheel)	**Shifting pattern on the cluster**. Instrument cluster.	None.	Feel the movement.
Mercedes GLA45 4Matic (Monostable)	Shifting pattern is shown. Red for the chosen gear. **Shifting pattern on the instrument cluster**.	Mechanical sound from the paddles for M.	Press down a button for P. Paddles for M.
Mesarati Quattroporte and Ghibli 2014 (monostable)	Shifting pattern is always shown. Higher intensity for the chosen gear. Instrument cluster.	Mechanical sound from pedals (M).	Feel the paddles in M. Feel the direction in which the shifter moves.

Shifter name (car model)	Image
ZF E-Shifter Fiat Chrysler: Dodge Charger, Chrysler 300, Jeep Grand Cherokee	  Image source: The Fast Lane Car (2016, 22 June). *Jeep's Recalled Gearshift Issue Demonstrated, Examined & Explained* [video file]. https://www.youtube.com/watch?v=AC6Uldx1mi8
Steptronic (Monostable R, N, D) BMW 5-series	  Image source: Luther Park Place Motor Cars (2011, 15 December). *2012 | BMW | 5 Series | Steptronic Shifting Functions | How To Luther Park Place Motors Rochester MN* [video file]. https://www.youtube.com/watch?v=EGy7lnZ7AIM
DCT BMW M5 (Monostable)	  Image source: Kyle McConell. (2013, 16 February). *MotiveMods M5 Shifter Retrofit Kit 135i E82* [video file]. https://www.youtube.com/watch?v=CbqnfYu-gek

Shifter name (car model)	Image
2016 Acura TLX (Button Shifter)	  Image source: OfficialAcuraCanada (2014, 29 September). *How to Use Acura's Electronic Gear Selector* [video file]. https://www.youtube.com/watch?v=TH5YYBIcUEE
Lincoln MKZ (Button Shifter)	  Image source: LincolnOwner (2013, 19 February). *Push-Button Shift | Lincoln How-to Video* [video file]. https://www.youtube.com/watch?v=bGZCTly-rkc
Nissan Leaf (Monostable)	  Image source: Dustin Batchelor (2013, 16 November). *2012 Nissan Leaf SL - How to turn on, drive / reverse, and display cluster explained* [video file]. https://www.youtube.com/watch?v=LyDJUwv8FFk

Shifter name (car model)	Image
BMW i3 (Paddle on the wheel)	  Image source: Cristian Pop (2016, 21 September). *Drive test bmw i3 gear shift and parking [EVmagazine.ro]* [video file]. https://www.youtube.com/watch?v=fxdVMB9YNXA
7G-DCT. Mercedes Benz. (Paddle on the wheel)	  Image source: Turayii (2014, 16 October). *Mercedes-Benz A180 CDI 7G-DCT explicit video 2 of 3* [video file]. https://www.youtube.com/watch?v=4E-gdElQDXA
Mesarati Quattroporte and Ghibli 2014 (monostable)	  Image source: rpo305 (2015, 10 November). *Shifting Your Maserati Ghibli(Sport/Manual)* [video file]. https://www.youtube.com/watch?v=Oq3_Zp19pP4

APPENDIX M – FLOWCHART FOR THE SHIFTER PROTOTYPE



APPENDIX N – WORD CLOUDS FROM THE PRODUCT REACTION CARDS

Concept C:



Concept V:



Concept A:



Concept VT:



Concept VA:


Distraherande
Begriplig
Pålitlig
Trygg
Förutsägbar
Lugn
Störande
Tidssparande
Hjälpsam
Stabil
Effektiv
Konsekvent
Irriterande
Enkel
EnkelAttAnvända

APPENDIX O – TASK COMPLETION TIME (SPSS DATA OUTPUT)

General Linear Model

Descriptive Statistics

	Mean	Std. Deviation	N
C	3.1644	.52636	25
V	2.6276	.47725	25
A	3.3159	.87215	25
VT	3.0771	.72474	25
VA	2.6861	.47155	25

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b]
					Greenhouse-Geisser
Time	.452	17.798	9	.038	.750

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.[a]

a. Design: Intercept

Within Subjects Design: Time

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
Time	Sphericity Assumed	9.166	4	2.292	12.233	.000
	Greenhouse-Geisser	9.166	3.000	3.056	12.233	.000
	Huynh-Feldt	9.166	3.476	2.637	12.233	.000
	Lower-bound	9.166	1.000	9.166	12.233	.002
Error(Time)	Sphericity Assumed	17.983	96	.187		
	Greenhouse-Geisser	17.983	71.996	.250		
	Huynh-Feldt	17.983	83.423	.216		
	Lower-bound	17.983	24.000	.749		

Estimated Marginal Means

Estimates

Measure: MEASURE_1

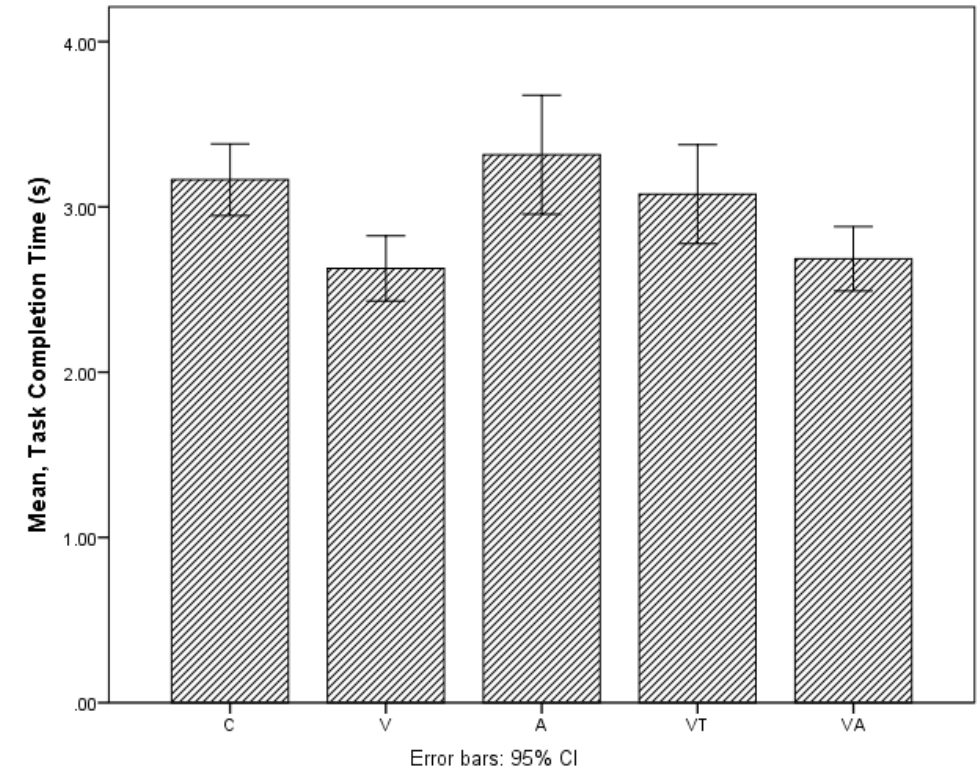
Time	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	3.164	.105	2.947	3.382
2	2.628	.095	2.431	2.825
3	3.316	.174	2.956	3.676
4	3.077	.145	2.778	3.376
5	2.686	.094	2.491	2.881

Pairwise Comparisons

Measure: MEASURE_1

(I) Time	(J) Time	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	.537*	.082	.000	.284	.789
	3	-.151	.137	1.000	-.576	.273
	4	.087	.137	1.000	-.337	.512
	5	.478*	.093	.000	.192	.765
2	1	-.537*	.082	.000	-.789	-.284
	3	-.688*	.121	.000	-1.063	-.313
	4	-.449*	.127	.016	-.841	-.058
	5	-.059	.089	1.000	-.335	.218
3	1	.151	.137	1.000	-.273	.576
	2	.688*	.121	.000	.313	1.063
	4	.239	.145	1.000	-.211	.688
	5	.630*	.152	.004	.161	1.099
4	1	-.087	.137	1.000	-.512	.337
	2	.449*	.127	.016	.058	.841
	3	-.239	.145	1.000	-.688	.211
	5	.391*	.118	.030	.025	.756
5	1	-.478*	.093	.000	-.765	-.192
	2	.059	.089	1.000	-.218	.335
	3	-.630*	.152	.004	-1.099	-.161
	4	-.391*	.118	.030	-.756	-.025

Graph


Mean, Task Completion Time (s)
4.00
3.00
2.00
1.00
.00
C
V
A
VT
VA
Error bars: 95% CI

APPENDIX P – OFF-ROAD GLANCES (SPSS DATA OUTPUT)

General Linear Model

Descriptive Statistics

	Mean	Std. Deviation	N
C	1.9876	.61407	25
V	1.6336	.55472	25
A	1.3896	.80139	25
VT	2.0260	.65756	25
VA	1.2616	.67152	25

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b]
					Greenhouse-Geisser
glances	.406	20.226	9	.017	.781

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.[a]

a. Design: Intercept

Within Subjects Design: glances

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
glances	Sphericity Assumed	11.845	4	2.961	15.471	.000
	Greenhouse-Geisser	11.845	3.122	3.794	15.471	.000
	Huynh-Feldt	11.845	3.643	3.252	15.471	.000
	Lower-bound	11.845	1.000	11.845	15.471	.001
Error(glances)	Sphericity Assumed	18.376	96	.191		
	Greenhouse-Geisser	18.376	74.929	.245		
	Huynh-Feldt	18.376	87.423	.210		
	Lower-bound	18.376	24.000	.766		

Estimated Marginal Means

Estimates

Measure: MEASURE_1

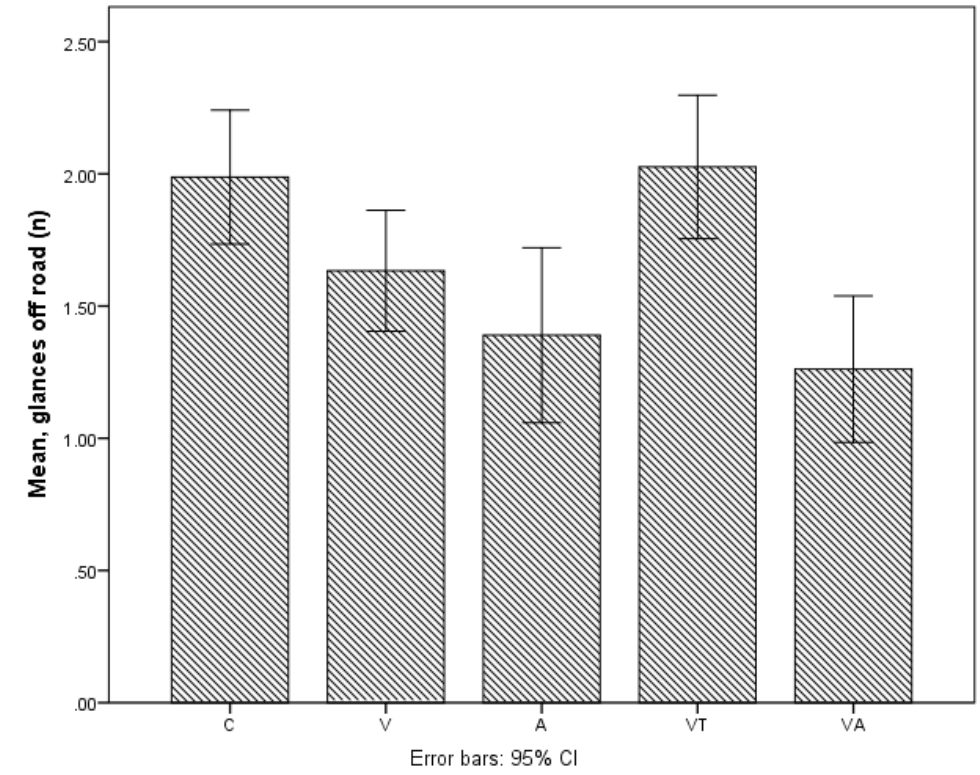
glances	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	1.988	.123	1.734	2.241
2	1.634	.111	1.405	1.863
3	1.390	.160	1.059	1.720
4	2.026	.132	1.755	2.297
5	1.262	.134	.984	1.539

Pairwise Comparisons

Measure: MEASURE_1

(I) glances	(J) glances	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	.354*	.074	.001	.124	.584
	3	.598*	.122	.001	.221	.975
	4	-.038	.154	1.000	-.513	.436
	5	.726*	.126	.000	.336	1.116
2	1	-.354*	.074	.001	-.584	-.124
	3	.244	.094	.159	-.047	.535
	4	-.392*	.121	.035	-.767	-.018
	5	.372*	.114	.032	.020	.724
3	1	-.598*	.122	.001	-.975	-.221
	2	-.244	.094	.159	-.535	.047
	4	-.636*	.147	.002	-1.091	-.182
	5	.128	.119	1.000	-.241	.497
4	1	.038	.154	1.000	-.436	.513
	2	.392*	.121	.035	.018	.767
	3	.636*	.147	.002	.182	1.091
	5	.764*	.144	.000	.318	1.211
5	1	-.726*	.126	.000	-1.116	-.336
	2	-.372*	.114	.032	-.724	-.020
	3	-.128	.119	1.000	-.497	.241
	4	-.764*	.144	.000	-1.211	-.318

Graph


Mean, glances off road (n)
2.50
2.00
1.50
1.00
.50
.00
C
V
A
VT
VA
Error bars: 95% CI

APPENDIX Q – NASA TLX (SPSS DATA OUTPUT)

General Linear Model

Descriptive Statistics

	Mean	Std. Deviation	N
C	60.0960	17.28623	25
V	34.7308	17.97904	25
A	49.7372	20.14180	25
VT	51.9452	18.00371	25
VA	30.3176	15.81042	25

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b]
					Greenhouse-Geisser
NASA	.369	22.333	9	.008	.699

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.[a]

a. Design: Intercept

Within Subjects Design: NASA

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
NASA	Sphericity Assumed	15473.200	4	3868.300	20.379	.000
	Greenhouse-Geisser	15473.200	2.798	5530.776	20.379	.000
	Huynh-Feldt	15473.200	3.204	4828.695	20.379	.000
	Lower-bound	15473.200	1.000	15473.200	20.379	.000
Error(NASA)	Sphericity Assumed	18222.723	96	189.820		
	Greenhouse-Geisser	18222.723	67.144	271.399		
	Huynh-Feldt	18222.723	76.906	236.947		
	Lower-bound	18222.723	24.000	759.280		

Estimated Marginal Means

Estimates

Measure: MEASURE_1

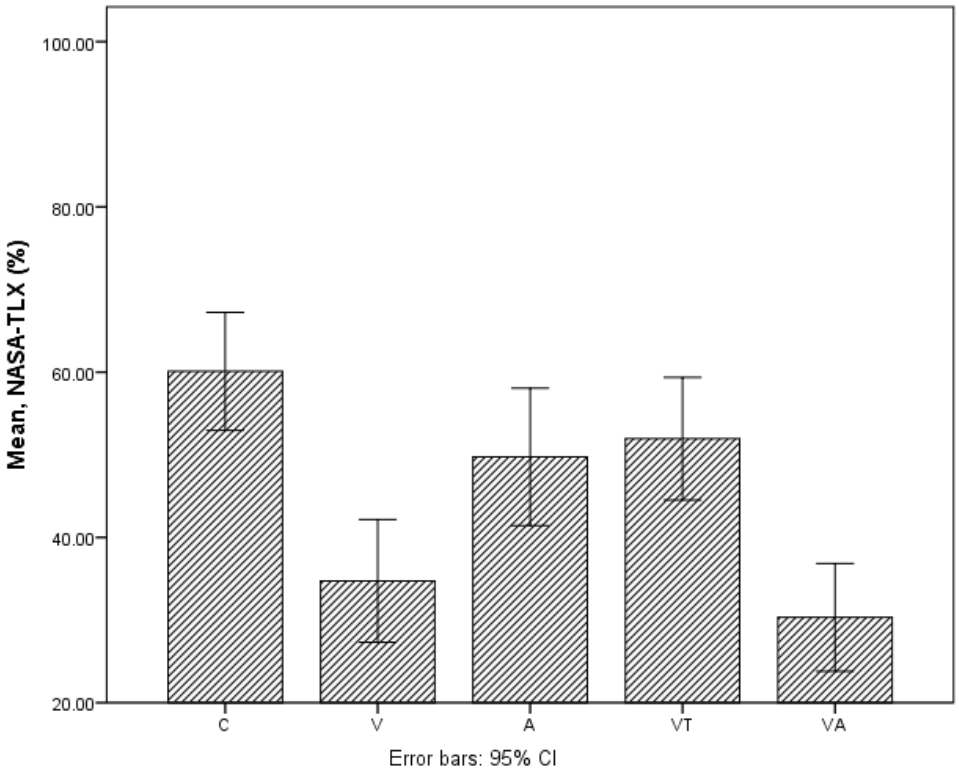
NASA	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	60.096	3.457	52.961	67.231
2	34.731	3.596	27.309	42.152
3	49.737	4.028	41.423	58.051
4	51.945	3.601	44.514	59.377
5	30.318	3.162	23.791	36.844

Pairwise Comparisons

Measure: MEASURE_1

(I) NASA	(J) NASA	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	25.365*	3.913	.000	13.274	37.457
	3	10.359	4.210	.214	-2.651	23.369
	4	8.151	2.786	.074	-.460	16.761
	5	29.778*	3.894	.000	17.745	41.812
2	1	-25.365*	3.913	.000	-37.457	-13.274
	3	-15.006*	4.057	.011	-27.543	-2.469
	4	-17.214*	4.400	.007	-30.811	-3.618
	5	4.413	2.319	.691	-2.754	11.580
3	1	-10.359	4.210	.214	-23.369	2.651
	2	15.006*	4.057	.011	2.469	27.543
	4	-2.208	4.242	1.000	-15.318	10.902
	5	19.420*	3.901	.000	7.365	31.474
4	1	-8.151	2.786	.074	-16.761	.460
	2	17.214*	4.400	.007	3.618	30.811
	3	2.208	4.242	1.000	-10.902	15.318
	5	21.628*	4.637	.001	7.296	35.959
5	1	-29.778*	3.894	.000	-41.812	-17.745
	2	-4.413	2.319	.691	-11.580	2.754
	3	-19.420*	3.901	.000	-31.474	-7.365
	4	-21.628*	4.637	.001	-35.959	-7.296

Graph


100.00
80.00
60.00
40.00
20.00
Mean, NASA-TLX (%)
C
V
A
VT
VA
Error bars: 95% CI

APPENDIX R – ACCEPTANCE SCALE - USEFULNESS (SPSS DATA OUTPUT)

General Linear Model

Descriptive Statistics

	Mean	Std. Deviation	N
C	-.4320	1.00942	25
V	1.0560	.51807	25
A	.5040	.87201	25
VT	.2960	.87392	25
VA	1.1680	.57353	25

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b]
					Greenhouse-Geisser
Useful	.358	23.036	9	.006	.698

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.[a]

a. Design: Intercept

Within Subjects Design: Useful

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
Useful	Sphericity Assumed	41.598	4	10.400	17.822	.000
	Greenhouse-Geisser	41.598	2.790	14.909	17.822	.000
	Huynh-Feldt	41.598	3.194	13.022	17.822	.000
	Lower-bound	41.598	1.000	41.598	17.822	.000
Error(Useful)	Sphericity Assumed	56.018	96	.584		
	Greenhouse-Geisser	56.018	66.963	.837		
	Huynh-Feldt	56.018	76.666	.731		
	Lower-bound	56.018	24.000	2.334		

Estimated Marginal Means

Estimates

Measure: MEASURE_1

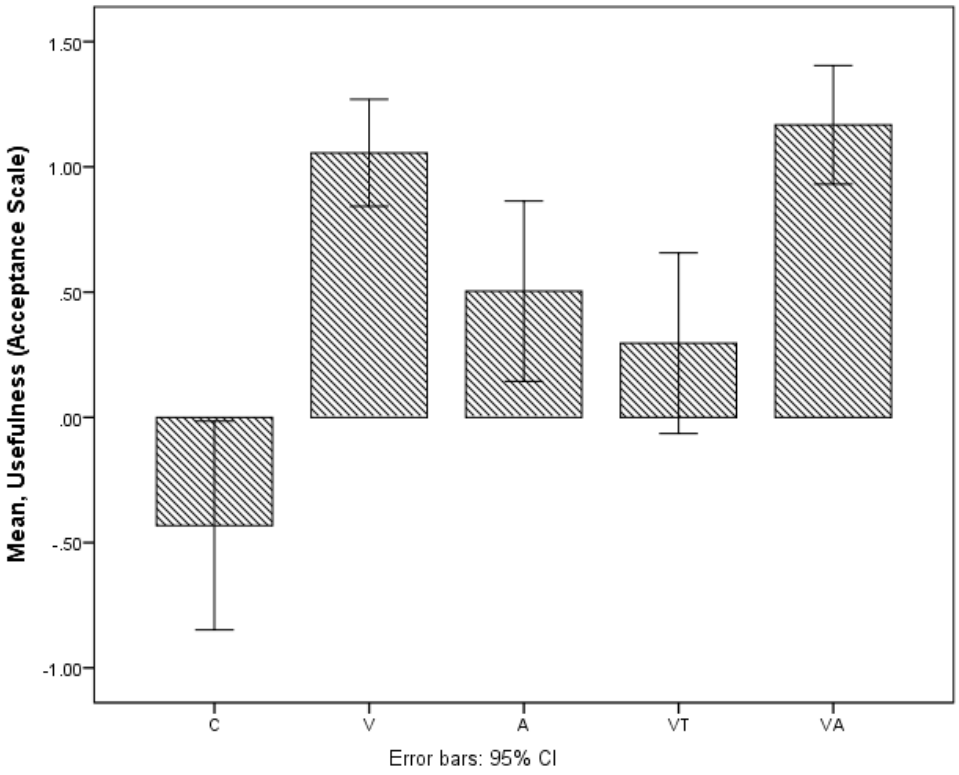
Useful	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	-.432	.202	-.849	-.015
2	1.056	.104	.842	1.270
3	.504	.174	.144	.864
4	.296	.175	-.065	.657
5	1.168	.115	.931	1.405

Pairwise Comparisons

Measure: MEASURE_1

(I) Useful	(J) Useful	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-1.488*	.171	.000	-2.016	-.960
	3	-.936*	.267	.018	-1.760	-.112
	4	-.728	.250	.076	-1.501	.045
	5	-1.600*	.280	.000	-2.464	-.736
2	1	1.488*	.171	.000	.960	2.016
	3	.552	.197	.100	-.058	1.162
	4	.760*	.204	.011	.129	1.391
	5	-.112	.171	1.000	-.641	.417
3	1	.936*	.267	.018	.112	1.760
	2	-.552	.197	.100	-1.162	.058
	4	.208	.215	1.000	-.456	.872
	5	-.664*	.161	.004	-1.162	-.166
4	1	.728	.250	.076	-.045	1.501
	2	-.760*	.204	.011	-1.391	-.129
	3	-.208	.215	1.000	-.872	.456
	5	-.872*	.209	.003	-1.518	-.226
5	1	1.600*	.280	.000	.736	2.464
	2	.112	.171	1.000	-.417	.641
	3	.664*	.161	.004	.166	1.162
	4	.872*	.209	.003	.226	1.518

Graph


Mean, Usefulness (Acceptance Scale)
1.50
1.00
.50
.00
-.50
-1.00
C
V
A
VT
VA
Error bars: 95% CI

APPENDIX S – ACCEPTANCE SCALE - SATISFACTION (SPSS DATA PUTPUT)

General Linear Model

Descriptive Statistics

	Mean	Std. Deviation	N
C	-.3800	.91036	25
V	.7700	.51498	25
A	.1100	.69252	25
VT	.1000	.80039	25
VA	.2800	.77500	25

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b]
					Greenhouse-Geisser
Satisfaction	.388	21.205	9	.012	.634

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.[a]

a. Design: Intercept

Within Subjects Design: Satisfaction

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F
Satisfaction	Sphericity Assumed	17.073	4	4.268	8.290
	Greenhouse-Geisser	17.073	2.536	6.732	8.290
	Huynh-Feldt	17.073	2.861	5.968	8.290
	Lower-bound	17.073	1.000	17.073	8.290
Error(Satisfaction)	Sphericity Assumed	49.427	96	.515	
	Greenhouse-Geisser	49.427	60.867	.812	
	Huynh-Feldt	49.427	68.659	.720	
	Lower-bound	49.427	24.000	2.059	

Estimated Marginal Means

Estimates

Measure: MEASURE_1

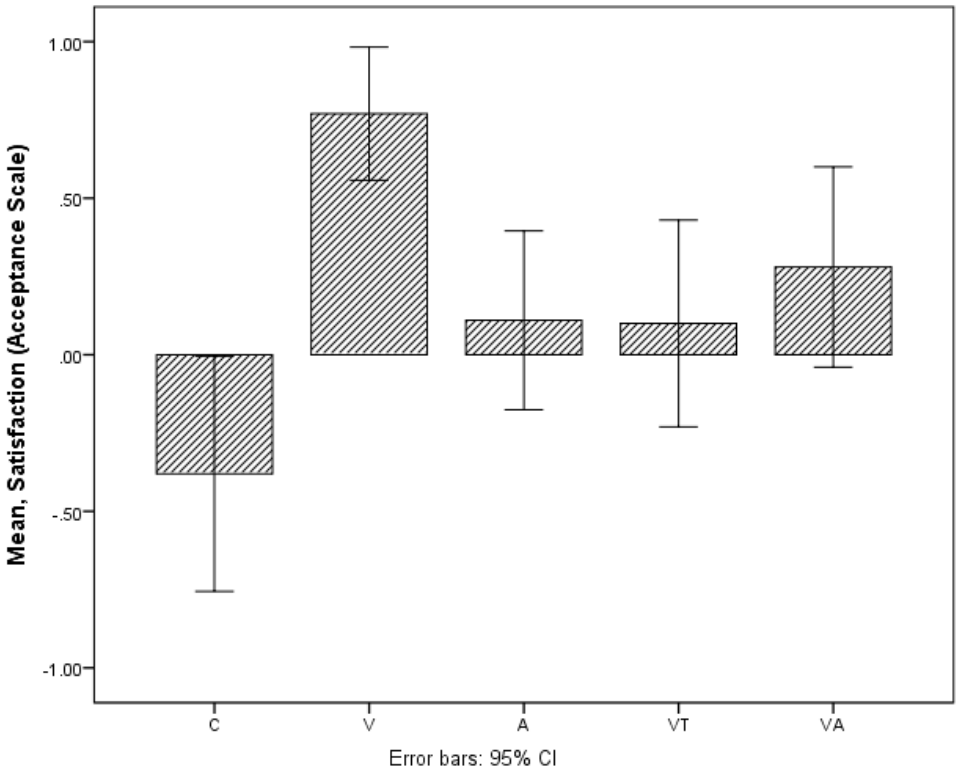
Satisfaction	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	-.380	.182	-.756	-.004
2	.770	.103	.557	.983
3	.110	.139	-.176	.396
4	.100	.160	-.230	.430
5	.280	.155	-.040	.600

Pairwise Comparisons

Measure: MEASURE_1

(I) Satisfaction	(J) Satisfaction	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-1.150*	.193	.000	-1.745	-.555
	3	-.490	.227	.408	-1.190	.210
	4	-.480	.165	.075	-.988	.028
	5	-.660	.280	.268	-1.525	.205
2	1	1.150*	.193	.000	.555	1.745
	3	.660*	.159	.004	.168	1.152
	4	.670*	.186	.014	.096	1.244
	5	.490	.181	.124	-.070	1.050
3	1	.490	.227	.408	-.210	1.190
	2	-.660*	.159	.004	-1.152	-.168
	4	.010	.187	1.000	-.567	.587
	5	-.170	.165	1.000	-.680	.340
4	1	.480	.165	.075	-.028	.988
	2	-.670*	.186	.014	-1.244	-.096
	3	-.010	.187	1.000	-.587	.567
	5	-.180	.251	1.000	-.957	.597
5	1	.660	.280	.268	-.205	1.525
	2	-.490	.181	.124	-1.050	.070
	3	.170	.165	1.000	-.340	.680
	4	.180	.251	1.000	-.597	.957

Graph


Mean, Satisfaction (Acceptance Scale)
1.00
.50
.00
-.50
-1.00
C
V
A
VT
VA
Error bars: 95% CI

APPENDIX S – RANKING (SPSS DATA OUTPUT)

General Linear Model

Descriptive Statistics

	Mean	Std. Deviation	N
C	1.48	.963	25
V	4.12	1.013	25
A	2.76	.926	25
VT	2.84	1.106	25
VA	3.80	1.384	25

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b]
					Greenhouse-Geisser
Rank	.232	32.723	9	.000	.604

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.[a]

a. Design: Intercept

Within Subjects Design: Rank

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.
Rank	Sphericity Assumed	107.200	4	26.800	18.017	.000
	Greenhouse-Geisser	107.200	2.418	44.336	18.017	.000
	Huynh-Feldt	107.200	2.708	39.584	18.017	.000
	Lower-bound	107.200	1.000	107.200	18.017	.000
Error(Rank)	Sphericity Assumed	142.800	96	1.488		
	Greenhouse-Geisser	142.800	58.030	2.461		
	Huynh-Feldt	142.800	64.996	2.197		
	Lower-bound	142.800	24.000	5.950		

Estimated Marginal Means

Estimates

Measure: MEASURE_1

Rank	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	1.480	.193	1.083	1.877
2	4.120	.203	3.702	4.538
3	2.760	.185	2.378	3.142
4	2.840	.221	2.383	3.297
5	3.800	.277	3.229	4.371

Pairwise Comparisons

Measure: MEASURE_1

(I) Rank	(J) Rank	Mean Difference (I-J)	Std. Error	Sig.[b]	95% Confidence Interval for Difference[b]	
					Lower Bound	Upper Bound
1	2	-2.640*	.331	.000	-3.663	-1.617
	3	-1.280*	.280	.001	-2.145	-.415
	4	-1.360*	.230	.000	-2.071	-.649
	5	-2.320*	.427	.000	-3.640	-1.000
2	1	2.640*	.331	.000	1.617	3.663
	3	1.360*	.341	.005	.306	2.414
	4	1.280*	.274	.001	.433	2.127
	5	.320	.377	1.000	-.846	1.486
3	1	1.280*	.280	.001	.415	2.145
	2	-1.360*	.341	.005	-2.414	-.306
	4	-.080	.369	1.000	-1.221	1.061
	5	-1.040*	.280	.011	-1.905	-.175
4	1	1.360*	.230	.000	.649	2.071
	2	-1.280*	.274	.001	-2.127	-.433
	3	.080	.369	1.000	-1.061	1.221
	5	-.960	.467	.509	-2.404	.484
5	1	2.320*	.427	.000	1.000	3.640
	2	-.320	.377	1.000	-1.486	.846
	3	1.040*	.280	.011	.175	1.905
	4	.960	.467	.509	-.484	2.404

Graph

